

October 30, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

Swimming and flight

Prof. Donato Romano (donato.romano@santannapisa.it)



Exploration of underwater life with an acoustically controlled soft robotic fish

Robert K. Katzschmann*, Joseph DelPreto, Robert MacCurdy and Daniela Rus

INTRODUCTION

Problem addressed

Closeup and minimally disruptive observations of marine life are particularly useful when studying animals' behaviors, swim patterns, and interactions within their habitats (1, 2). A biomimetic underwater observatory for long-term studies could facilitate deeper understanding of marine life, especially their social behaviors and how environmental changes affect the delicate balance within the marine world. One possibility to achieve this is using underwater vehicles that can swim alongside marine life to allow closeup observations. Remotely operated vehicles (ROVs) or autonomous underwater vehicles (AUVs) in ocean environments typically use propellers or jet-based propulsion systems (3). However, these propulsion systems generate substantial turbulence and have the potential to scare marine life and to prevent closeup observations (4). Further, the mere appearance of these vehicles, typically large and rigid like a submarine, does not integrate well into the marine environment. The complexity of most traditional ROVs also requires costly fabrication and intricate control strategies, and their large bulk restricts their tethered deployment to deeper water using specially equipped vessels. Smaller ROVs also generally require tethers, which can be cumbersome and restrict operation.

To address this problem, we sought to create biomimetic fish robots that can be easily used by a single diver. We also wanted to generate lifelike undulation of the robotic fish tail for propulsion and to enable untethered remote control of these fish robots by a diver. Our solution is a soft underwater robot with fluid-driven actuation that swims with compliant and continuous strokes that imitate the movement of fish. Biomimicry potentially increases the ability of robots to approach marine life without disturbing them (5, 6) or their natural environment. Despite the emergence of previously unknown actuation modalities (7) that could also enable undulatory or oscillatory biomimetic locomotion underwater (8, 9), none of the bench-

This work presents (i) a powerful hydraulic soft actuator, (ii) a control mechanism that allows the robot to adjust its buoyancy according to depth, (iii) onboard sensors to observe and record the environment, (iv) a mission control system that a human diver can use to provide navigational commands to the robot from a distance using acoustic signals, and (v) extended ocean experiment to 18 m.

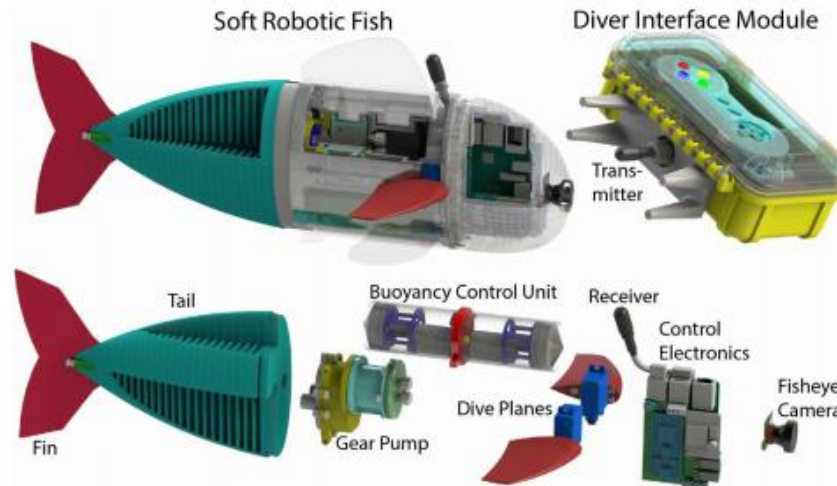


Fig. 1. SoFi system overview. (Top, left to right) Soft robotic fish and diver interface module. (Bottom, left to right) Subcomponents of the system are the elastomer tail (cut view), external gear pump, BCU, two dive planes, control electronics including acoustic receiver, and fisheye camera.



Fig. 2. Underwater exploration. Fish trajectory exploring a coral reef. The snapshots of the fish are equally spaced in time by 2.6 s per representation.

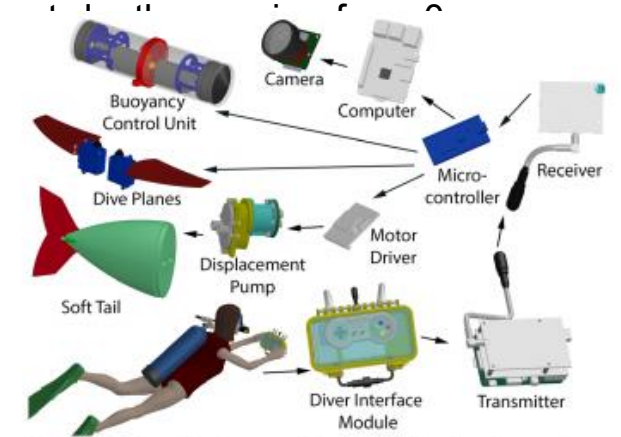


Fig. 6. Signal flow within the system. The command flow from a human diver to the robot. The diver sends acoustic commands such as thrust, left/right, and up/down as well as camera modes from the transmitter within the diver interface module. The analog signal travels several meters underwater and is then amplified by the receiver and parsed by the microcontroller. The microcontroller adjusts the pump speed, the dive plane position, the BCU, and the mode of the camera.

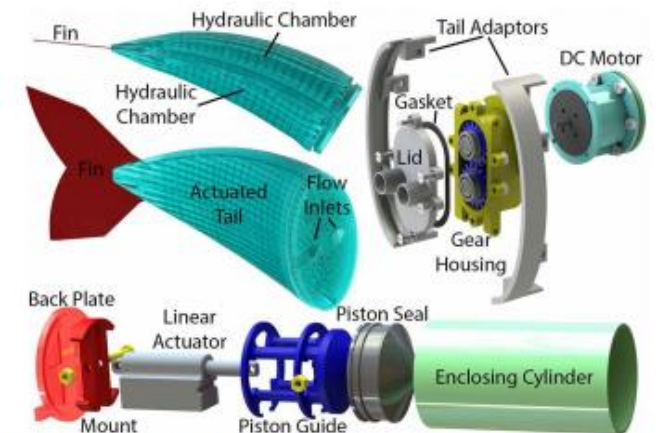


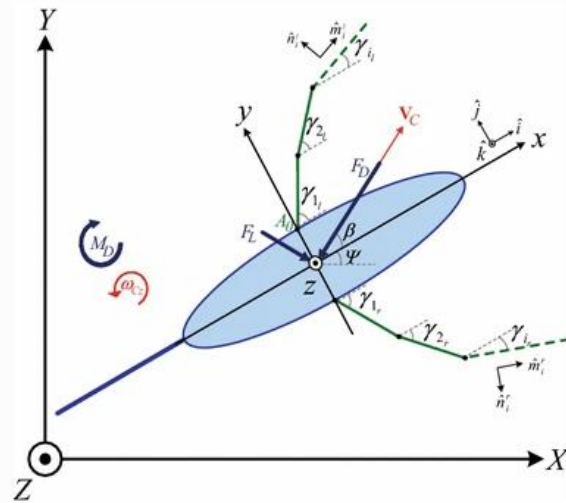
Fig. 7. Soft tail, pump, and BCU. (Top) Two views of the soft fish tail in an actuated state (left) and custom external gear pump in an exploded view (right). (Bottom) One of two identical BCU modules in an exploded view.

Improving maneuvering and locomotion of a robotic fish

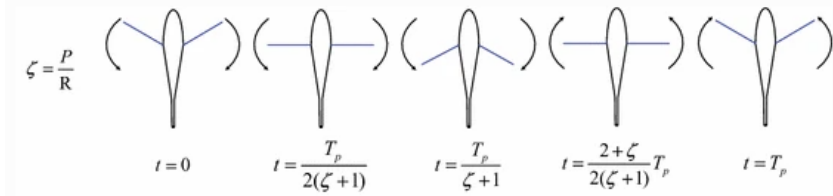
Improved efficiency, maneuverability, and stealth are some of the potential advantages of robotic fish over traditional propeller-driven underwater vehicles. The most challenging step in dynamically modeling robotic fish is capturing the interactions between the body/fins and the fluid and calculating the resulting force and moment exerted on the body.

Although the caudal fin is the primary appendage used for propulsion in robotic fish, pectoral fins play a crucial role in **maneuvering** and **stability** of fish, as well as providing or assisting propulsion.

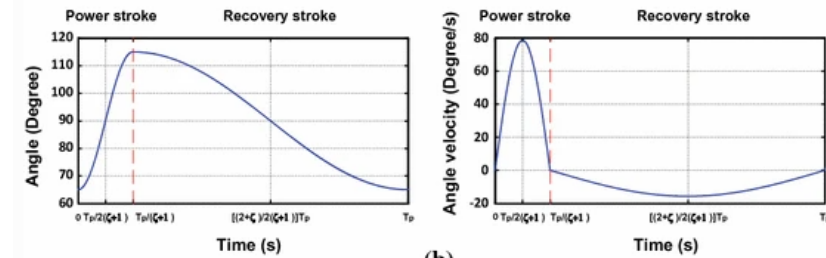
Behbahani & Tan (2017) developed a systematic, computation-efficient framework for analyzing how the flexibility of pectoral fins affects the swimming performance and mechanical efficiency of the robot.



Top view of the robotic fish actuated by flexible pectoral fins and restricted to planar motion



(a)

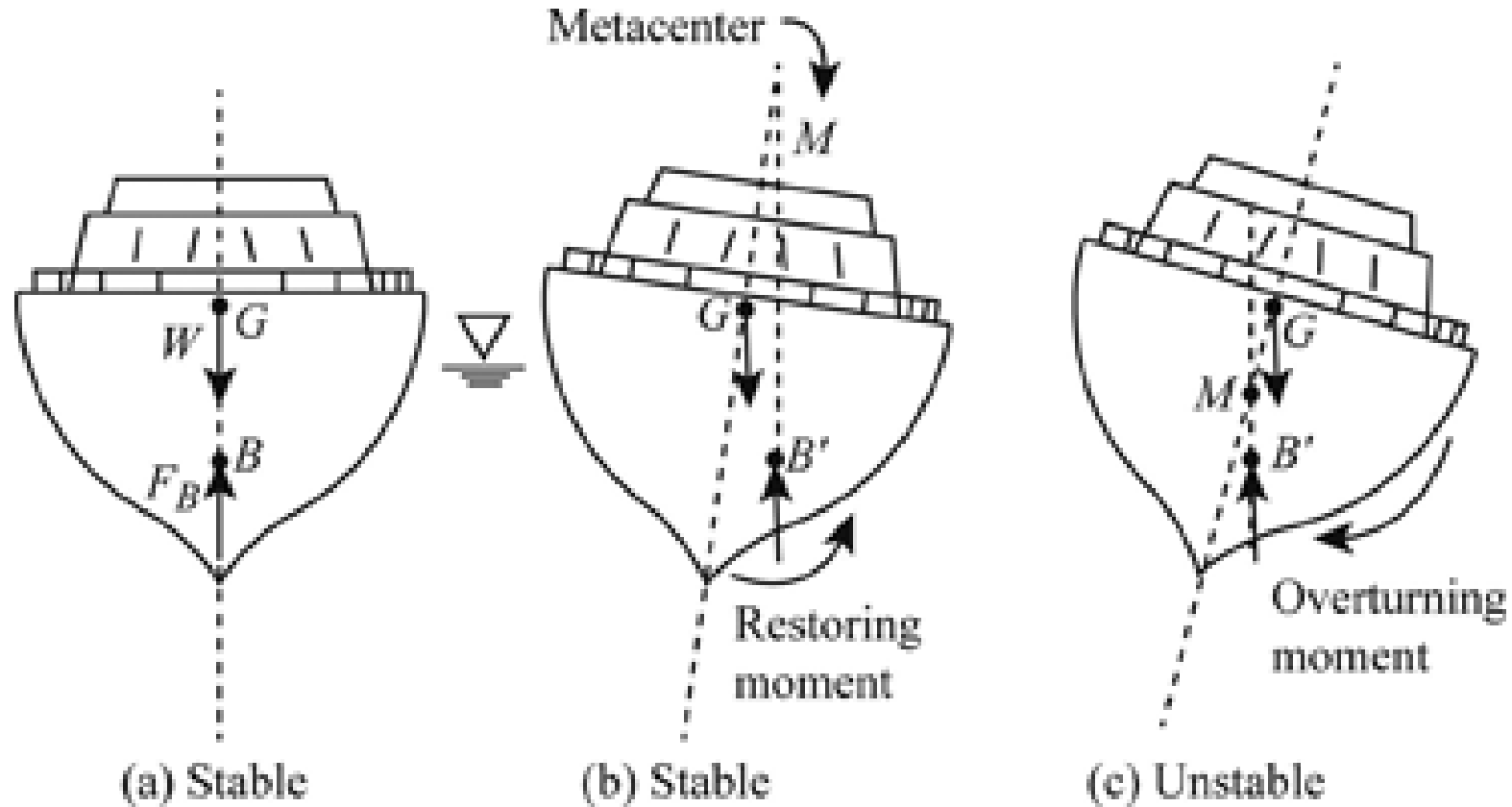


(b)

Illustration of the pectoral fin motion during power and recovery strokes: **a** the snapshots at different time instances of one fin-beat cycle, where T_p represents the total time for each cycle; **b** orientation and angular velocity of the base of the pectoral fin with respect to the main axis of the robotic fish, where $\zeta = 5$, $\gamma_{rMA} = 50$ deg, and $T_p = 1$ s are used

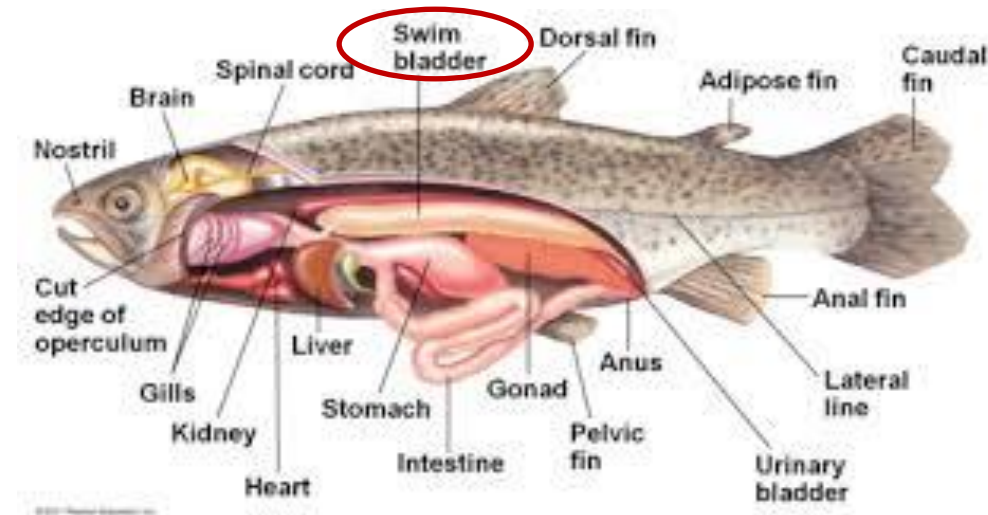
Behbahani, S. B., & Tan, X. (2017). Role of pectoral fin flexibility in robotic fish performance. *Journal of Nonlinear Science*, 27(4), 1155-1181.

Semi-Submerged (Floating) Objects



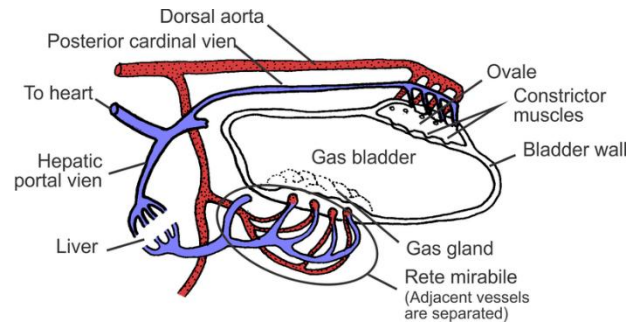
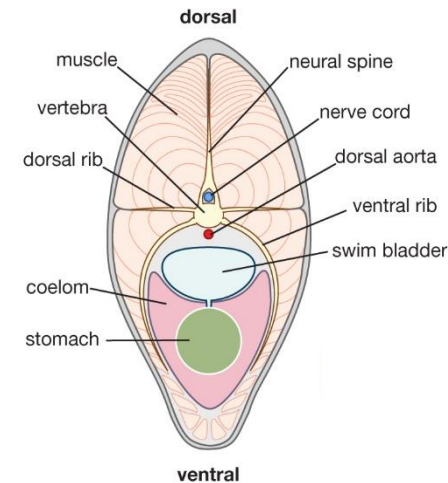
Buoyancy control

Bony fish have an internal gas-filled organ that contributes to their ability to control their buoyancy. This organ is named **swim bladder**, **gas bladder**, **fish maw**, or **air bladder**. Generally, the swim bladder includes two gas-filled sacs located in the dorsal portion of the fish. It has flexible walls that contract or expand according to the ambient pressure. The walls of the bladder contain very few blood vessels and are lined with guanine crystals, which make them impermeable to gases. By adjusting the gas pressurizing organ using the gas gland or oval window the fish can obtain neutral buoyancy and ascend and descend to a large range of depths.

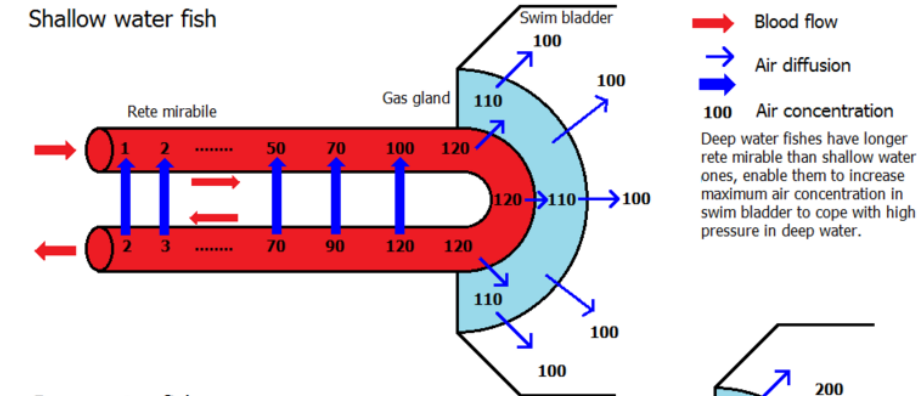


The dorsal position of the swim bladder means the **center of mass** is below the **center of volume**, allowing it to act as a stabilizing agent.

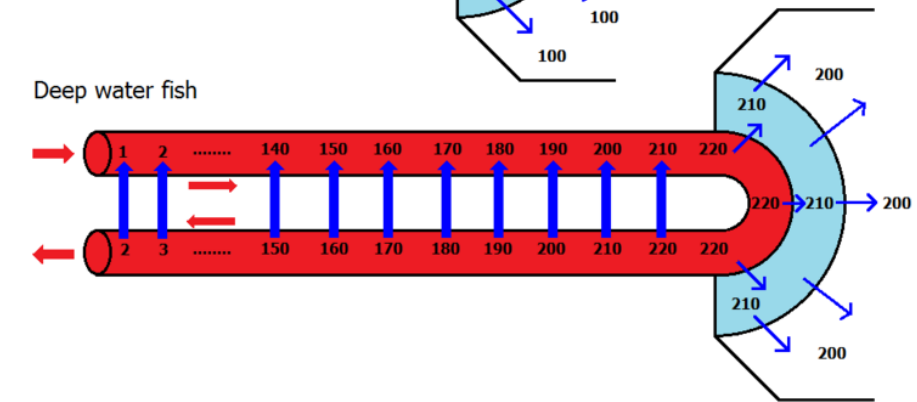
Teleost fish in cross section



Shallow water fish

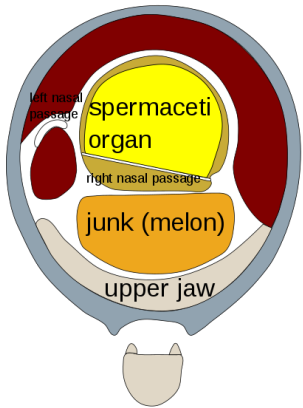


Deep water fish

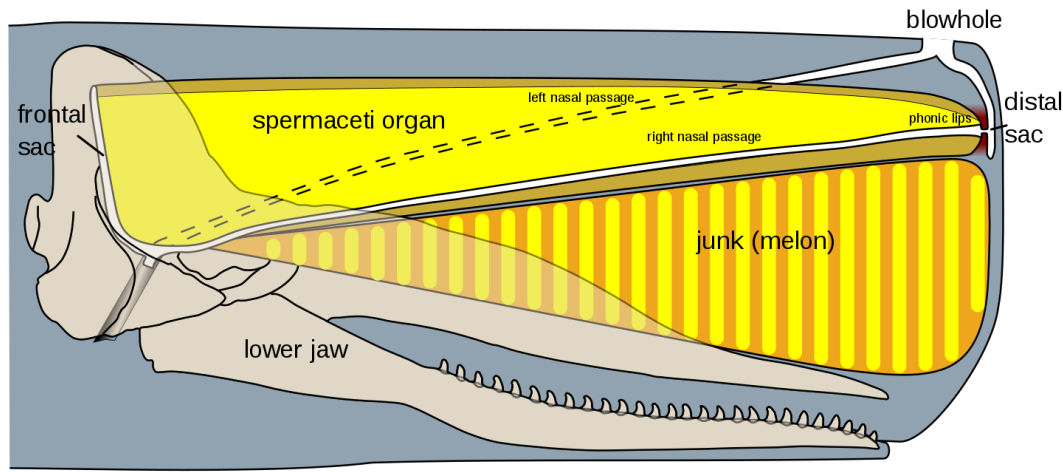


Buoyancy control

The **spermaceti organ** is an organ that is commonly present at level of the heads of toothed whales. This organ contains a liquid wax called spermaceti and is involved in the generation of sound. Furthermore, the spermaceti organ is involved in buoyancy manipulation via manipulation of the contained spermaceti oil's temperature and, in consequence, its density, facilitating deep diving by cooling and surfacing by warming, as well as allowing the animal to remain motionless at great depth.



TRANSVERSE



SAGITTAL

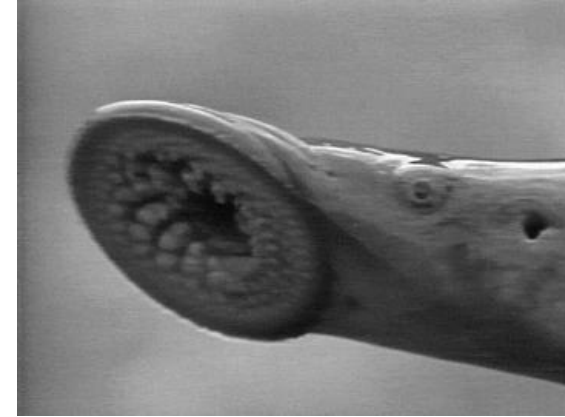


Lamprey as a Vertebrate Prototype

Lampreys

Phylum: Chordata

Class: Cephalaspidomorphi



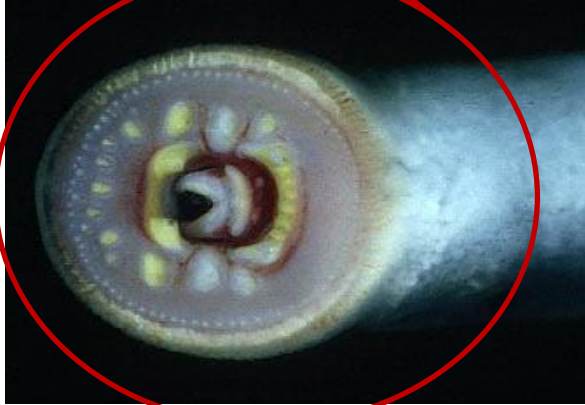
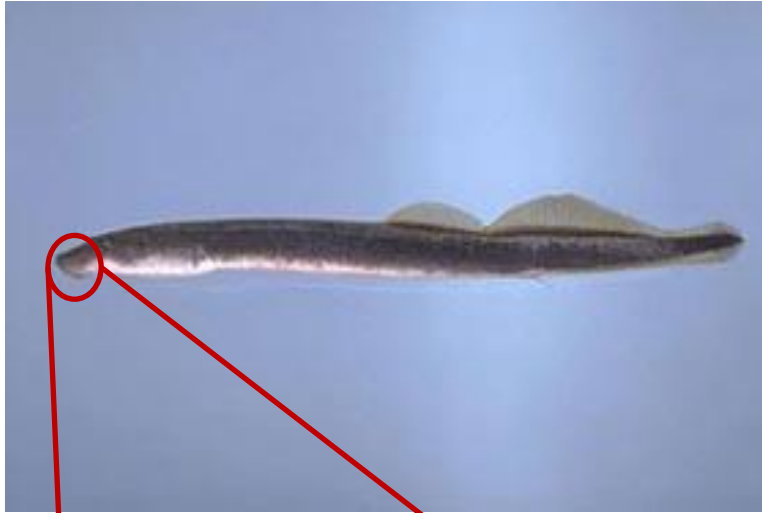
Considered the most basal group of the Vertebrata

- scaleless
- jawless
- lack paired fins

Lampreys generate thrust using lateral movements of their body and caudal fin



Lamprey as a Vertebrate Prototype



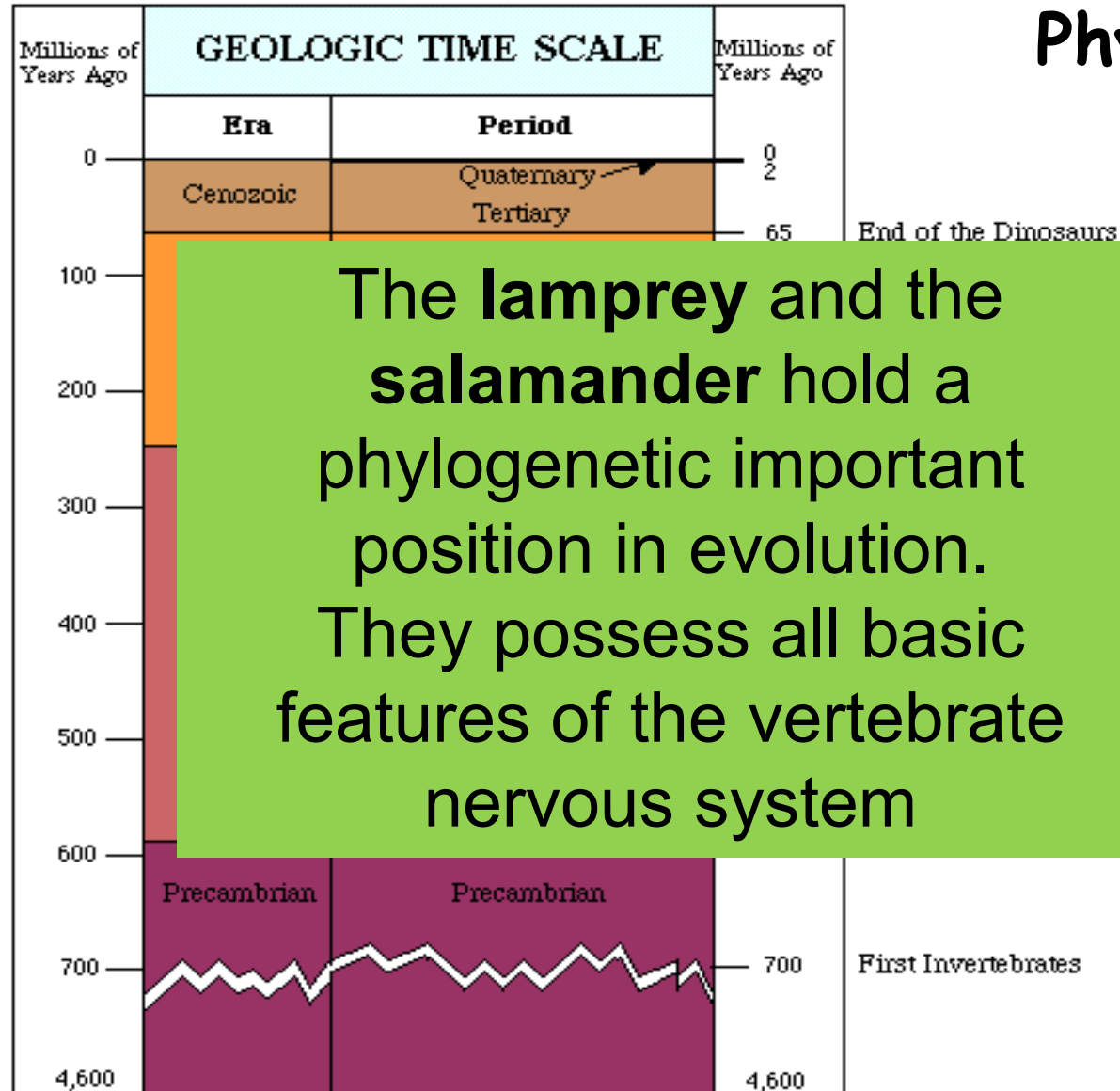
A phylogenetically important position in evolution

Diverged from the main vertebrate line 530 million years ago

Due to its relatively simple nervous system it represents an ideal biological model for bioinspiration

Developed all basic features of the vertebrate nervous system

Lampetra Strategy: early vertebrates as models to investigate



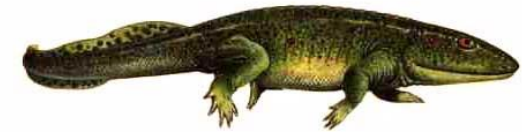
Phylogenetic tree

Salamander



Ichtyostega

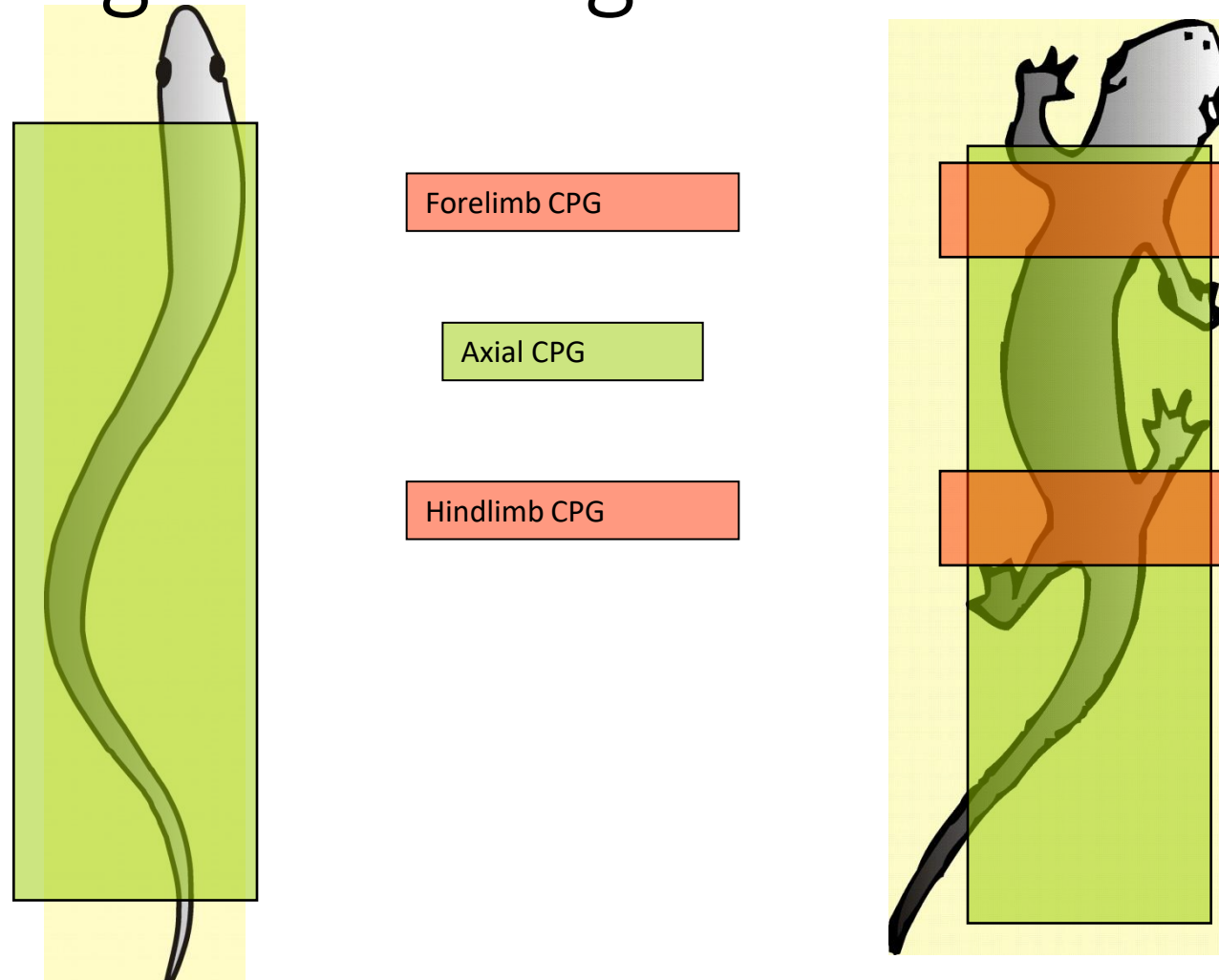
ls, Birds



Lamprey

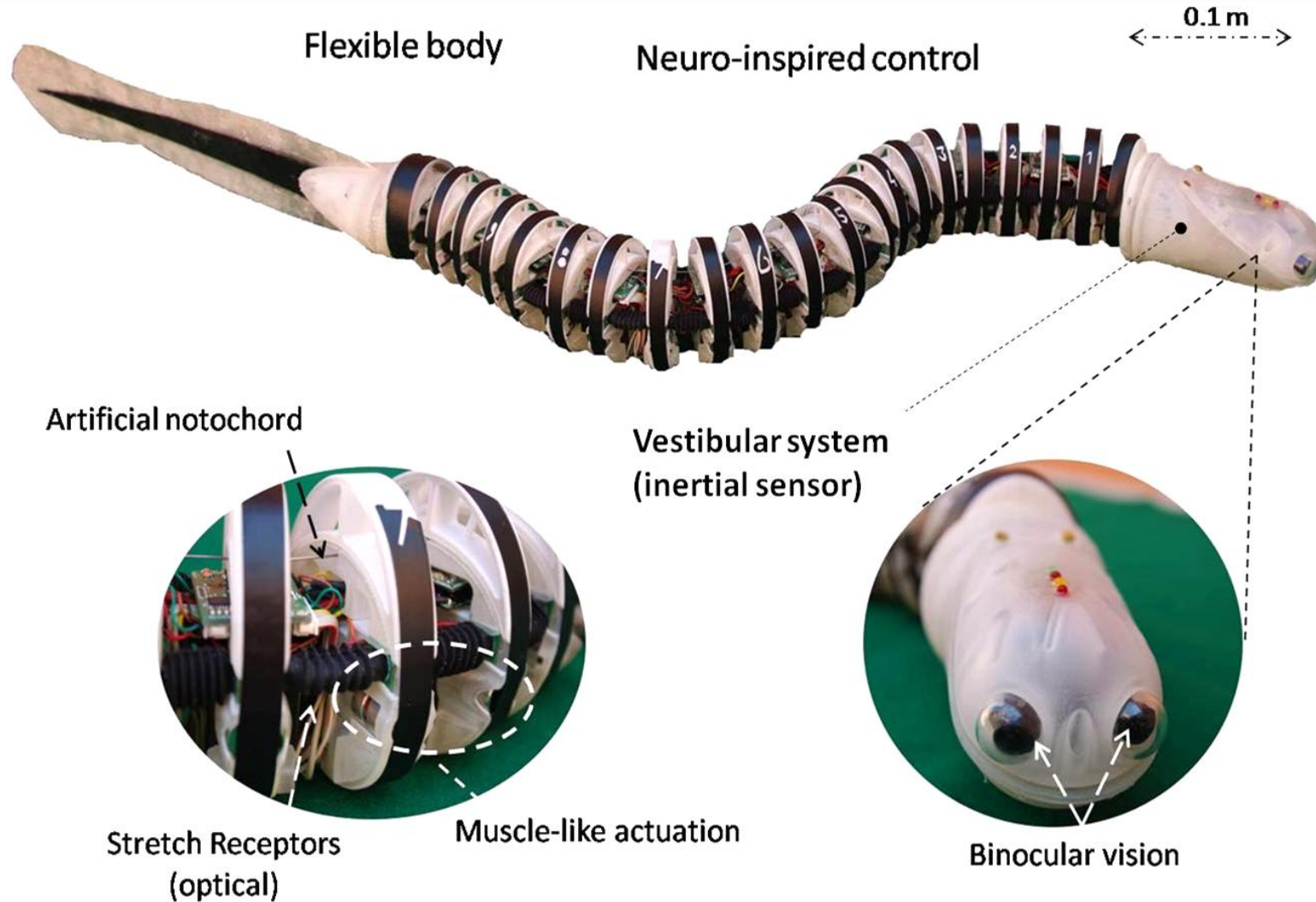


From swimming to walking...



Salamander CPG = Lamprey-like **axial CPG** extended with **2 limb CPGs**

The lamprey biomimetic artefact



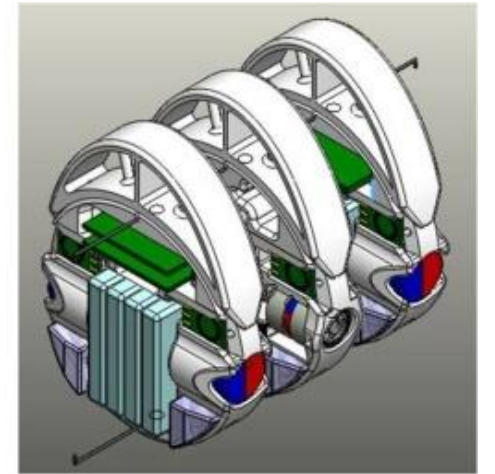
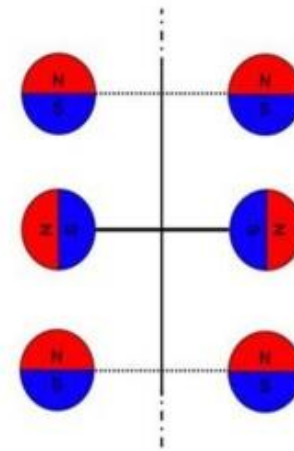
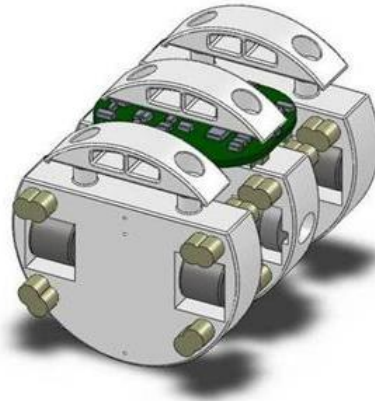
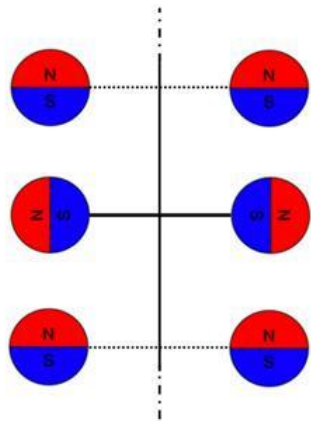
- COMPLIANT BODY
- 55 mm in diameter
- 860mm in length
- 10 motors
- 21 vertebrae
- Magnetic actuators (MUSCLE -LIKE)
- 10 battery packs
- Stretch sensors
- 2 2D cameras
- Wireless communication
- Gyros and accelerometers



Cesare Stefanini
The BioRobotics Institute
Scuola Superiore Sant'Anna and
Khalifa University, Abu Dhabi

Muscle-like actuation

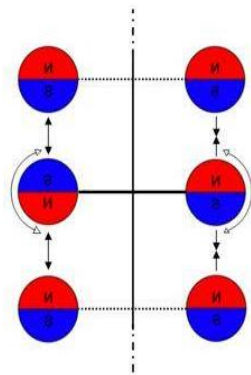
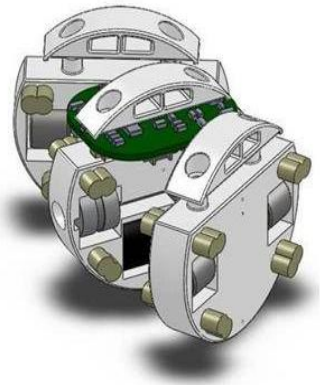
Neutral configuration



Neutral configuration is given by the balance of magnetic forces between left and right sides in the active and passive vertebrae

Muscle-like actuation

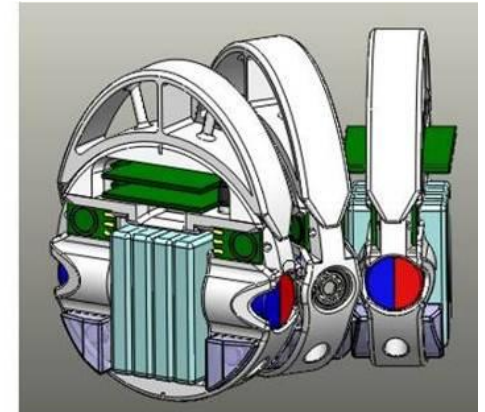
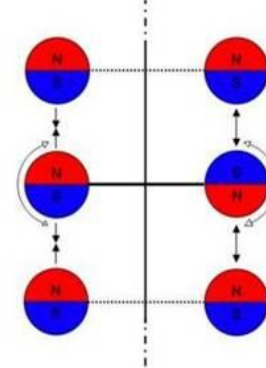
Bending on one side



Passive vertebra

Active vertebra

Passive vertebra



The magnet orientation is modified in order to produce a magnetic interaction between the vertebras (in this case the attractive interaction between the left side magnets and the repulsive interaction between right side magnets produces the left bending of the segment)

Experiments and tests

Colour tracking

Continue targeting



Spot targeting

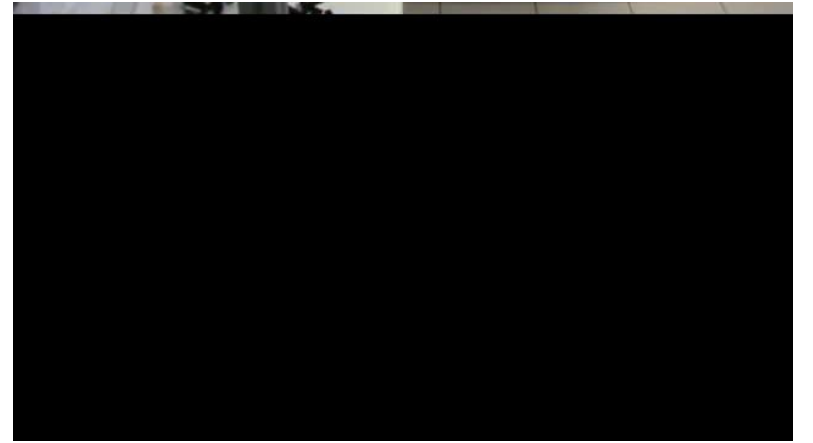


Experimental sessions **validating** basic sensorimotor mechanisms and **vision-driven locomotion** have been successfully performed **“in artefacto”**

Results published on Biological Cybernetics; n. 105; pp. 513:527 (2013)

locomotion

Light tracking



“Superquark” national TV program



“Lamprey Robot” Team

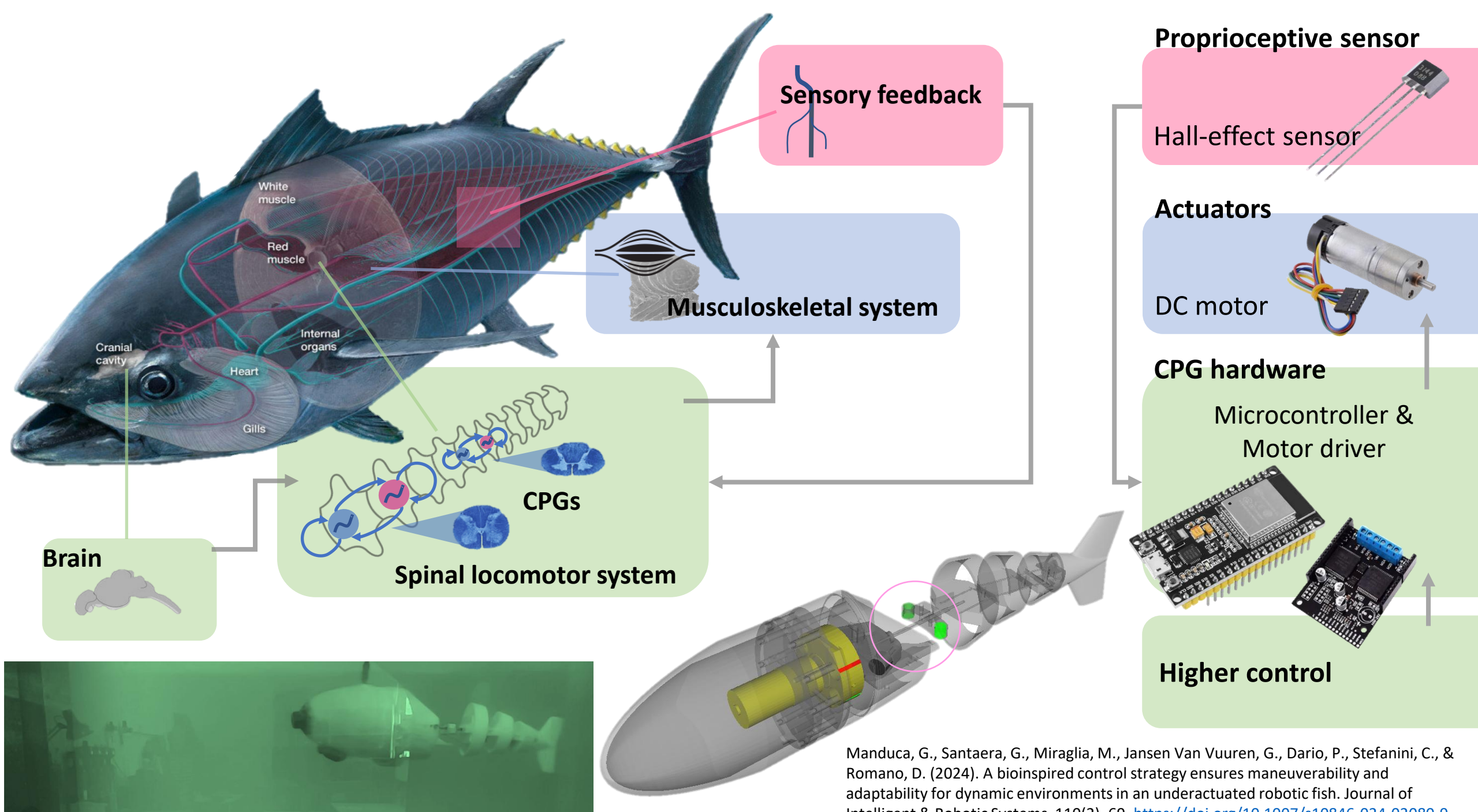


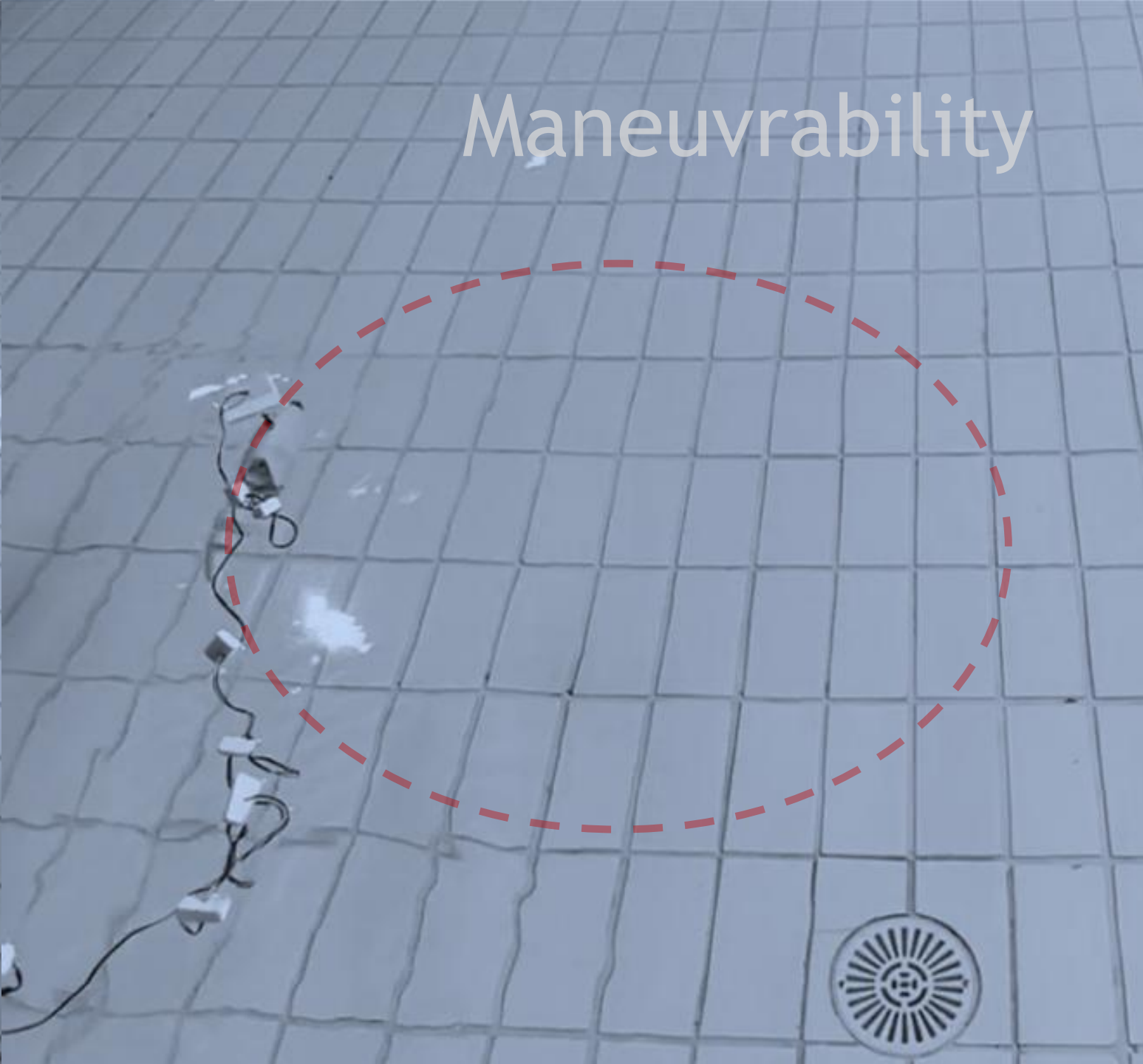
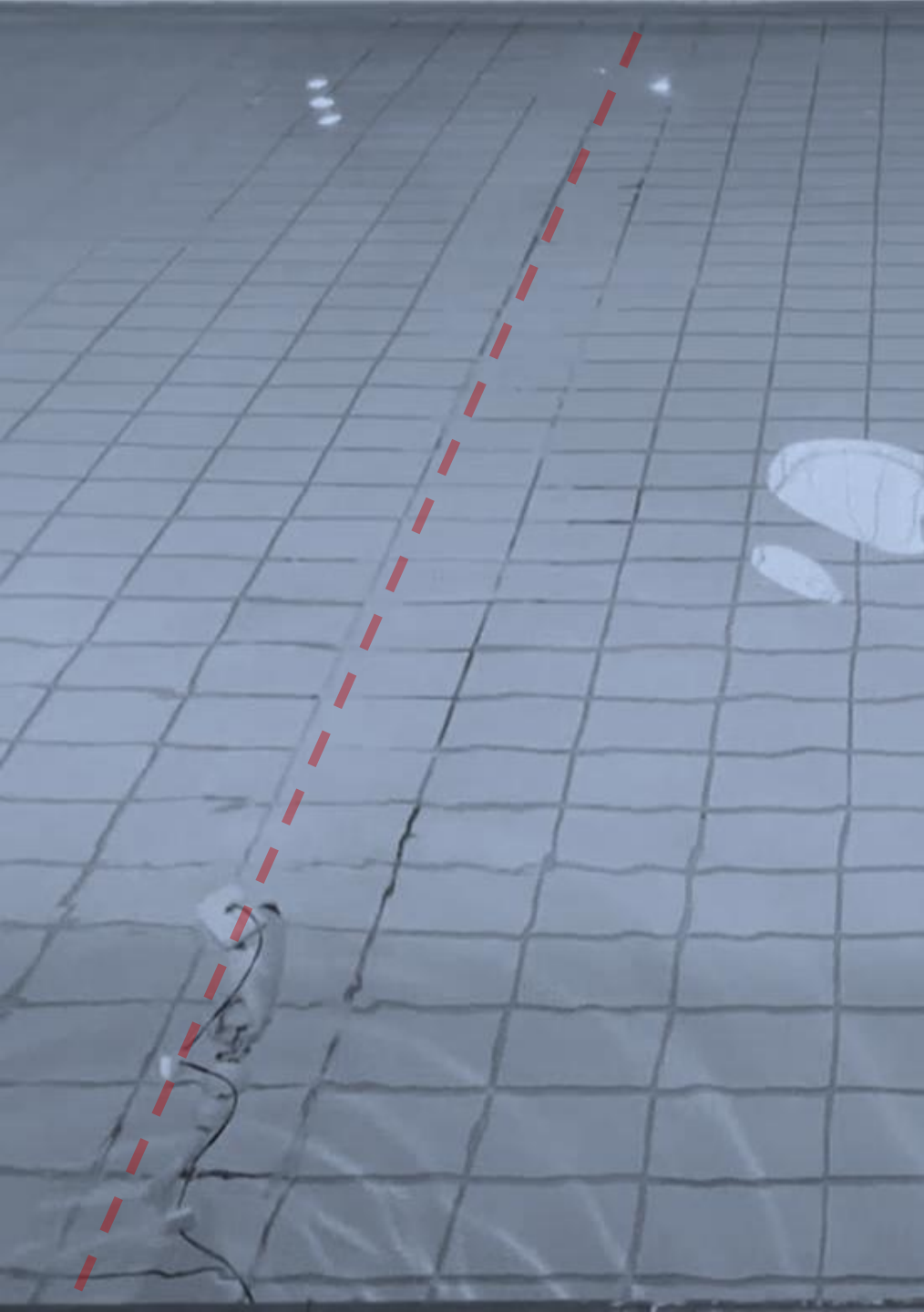


OCEAN Biodiversity



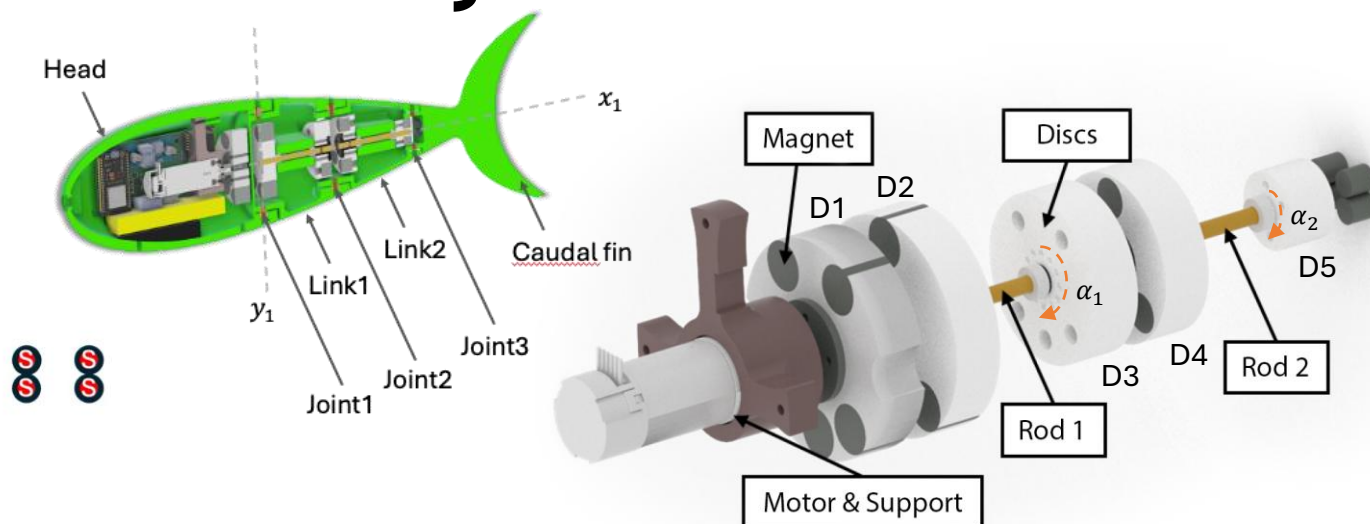
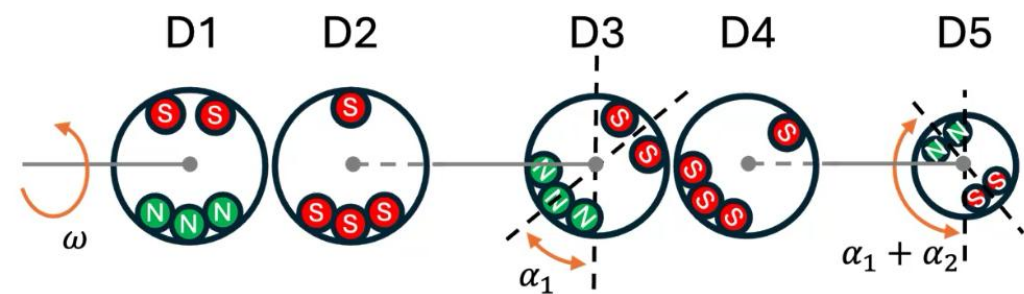
Aquafarming







A Fish robot for dynamic environments



Magnetically Transmitted Traveling Wave:

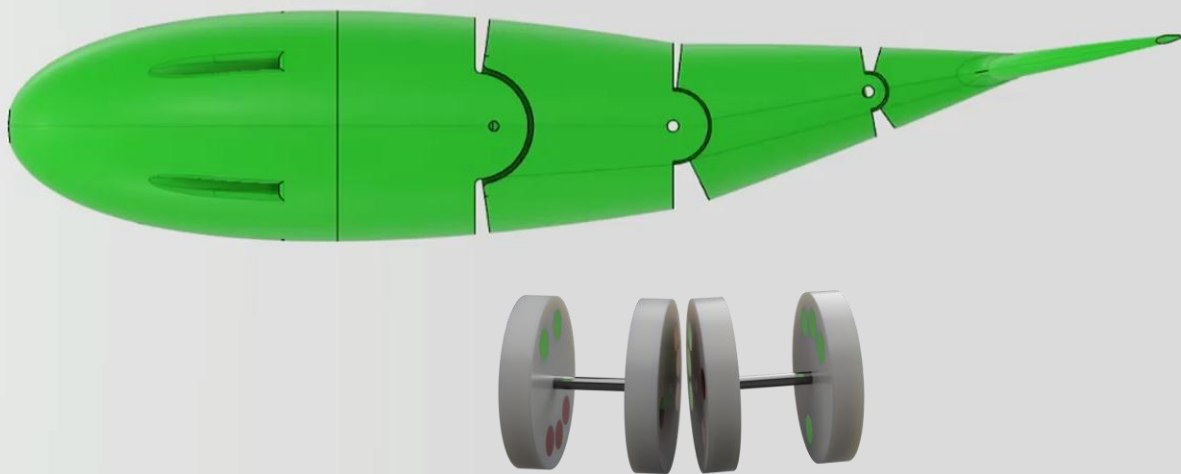
The mechanism transforms the unidirectional rotation of a single motor into an oscillatory, phase-shifted movement across the modules of the kinematic chain, generating a traveling wave along the body.

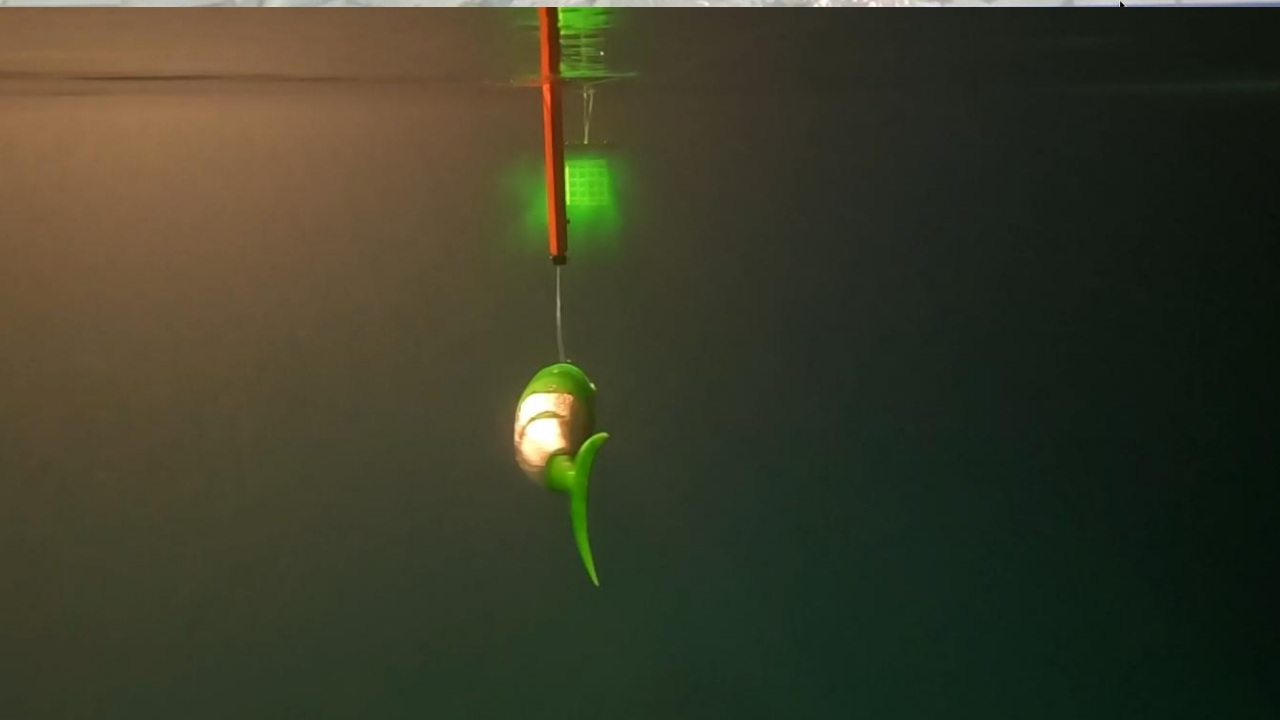
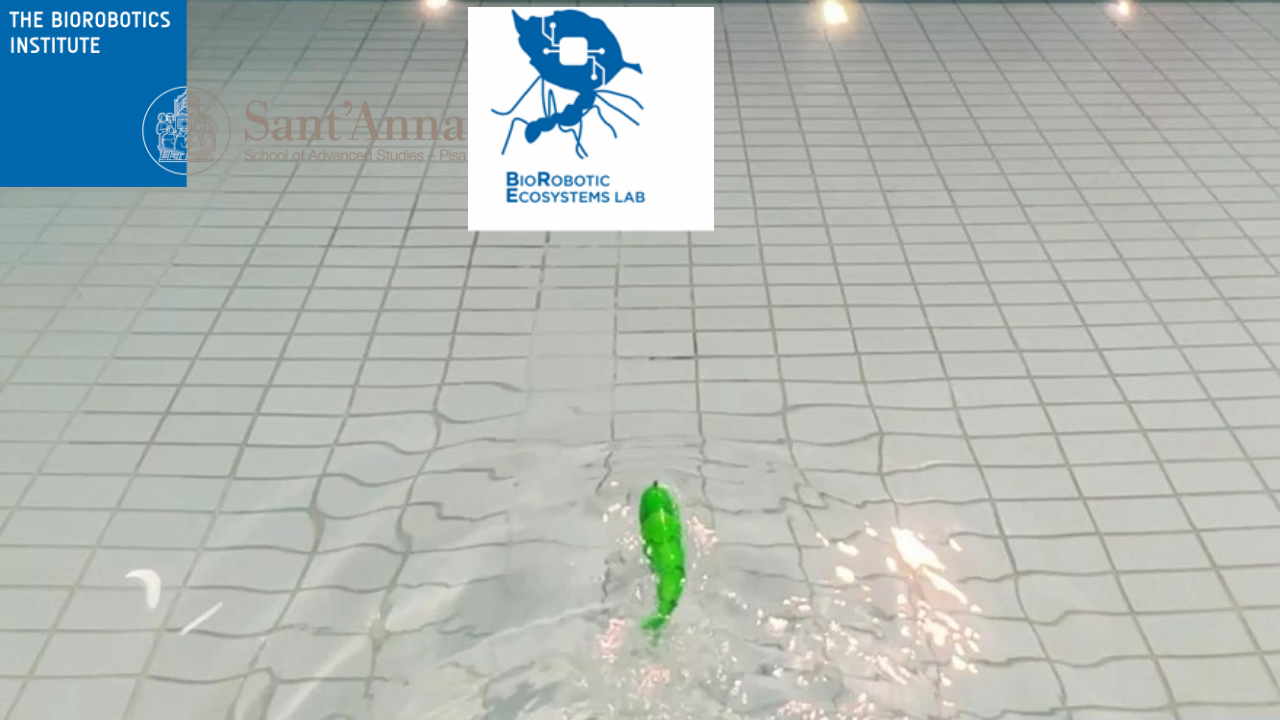
Advantages:

- **Propulsion efficiency** through traveling wave generation;
- **Underactuated** solution to improve the system reliability and energy savings;
- **Not a passive traveling wave generation**;
- The **non-blocking transmission system** can mitigate the risk of motor overload in case of tail obstruction;
- Ability to **isolate electronics**, thus enhancing waterproofing and hydrostatic pressure resistance;
- **Modular and scalable**.



Pat. App. no. PCT/IB2025/052691, Manduca, G., Padovani, L., Romano, D., & Stefanini, C. (2024). Underactuated magnetic transmission system.
Priority: 102024000005779. Priority date: 14 March 2024.





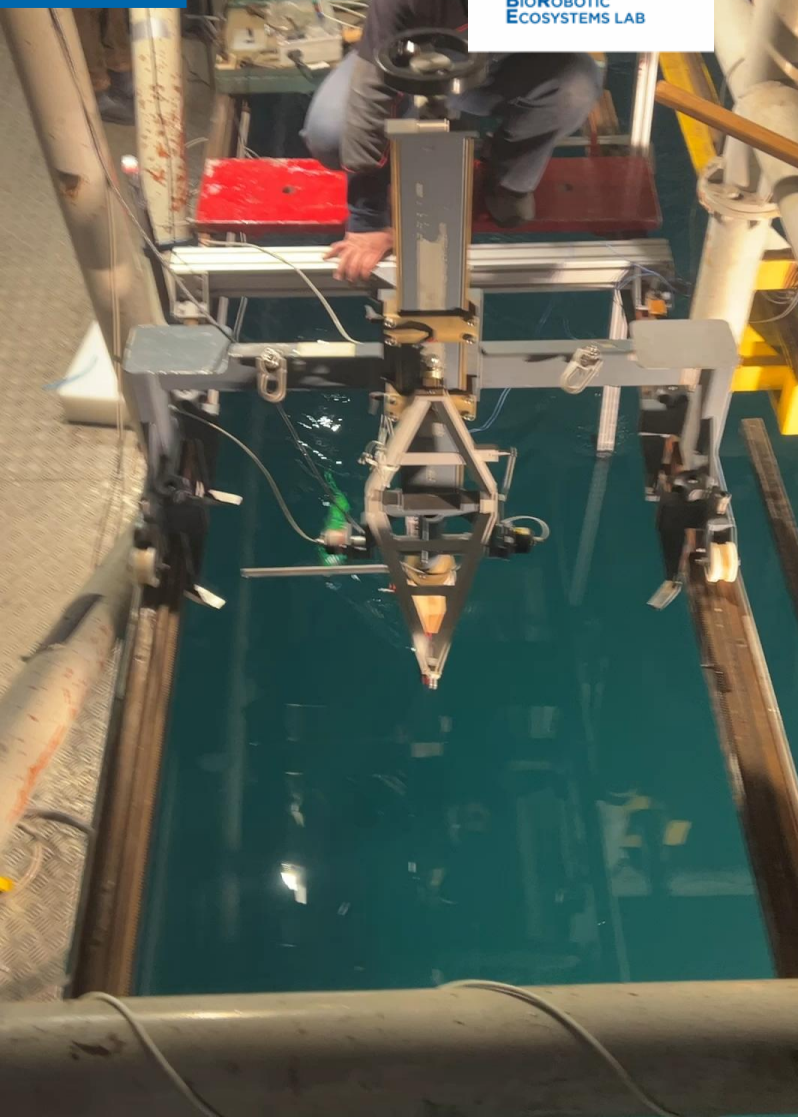
1 Hz

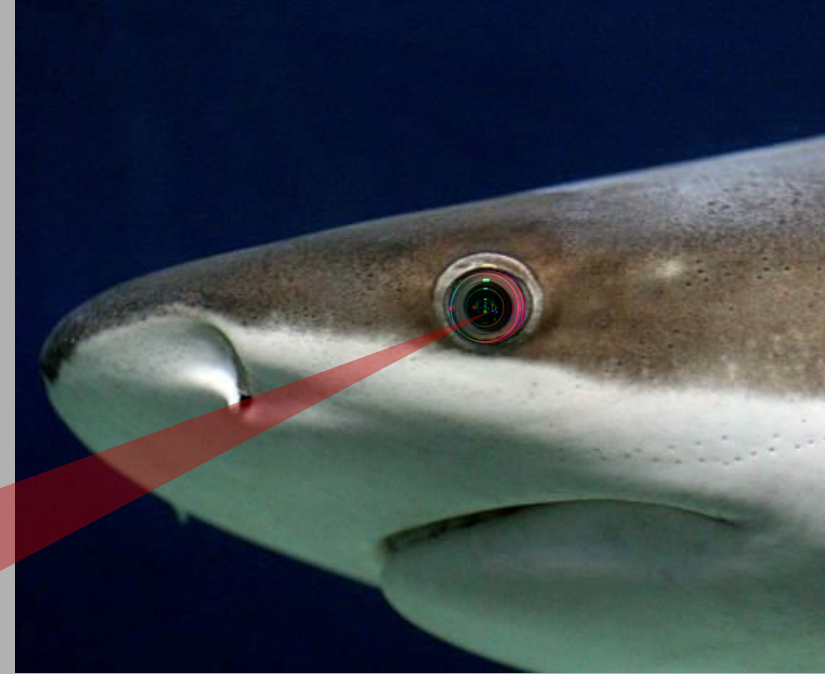
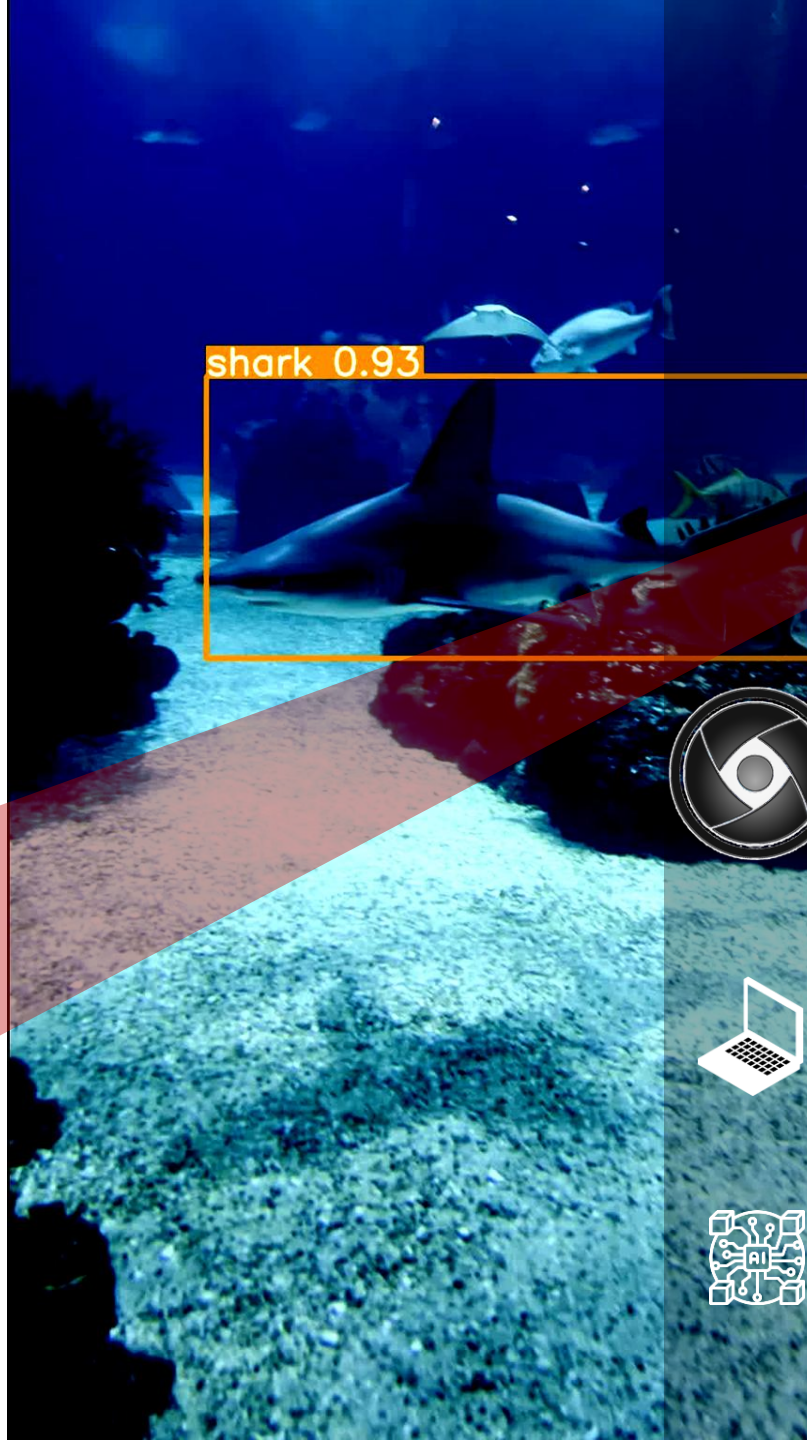
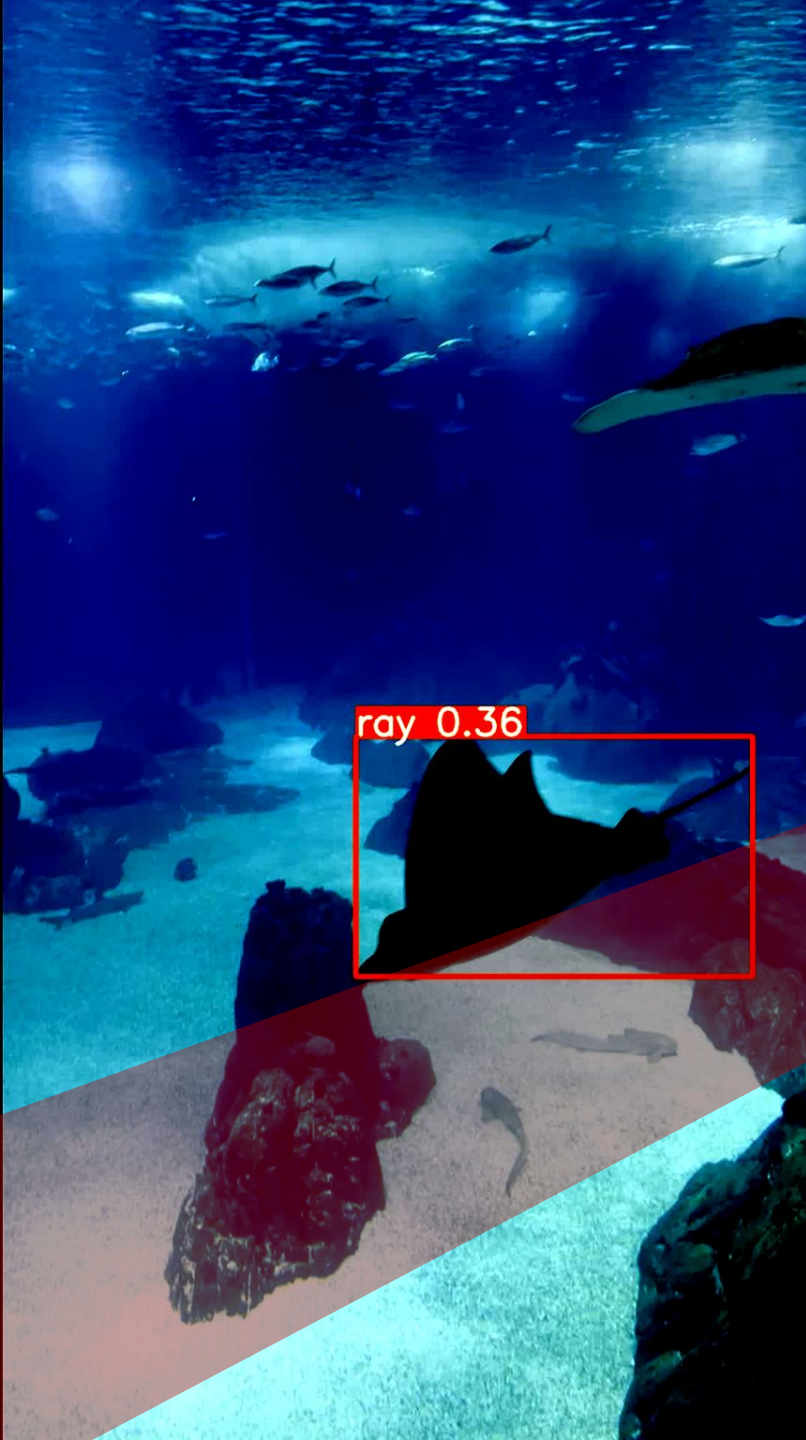


3 Hz

With a battery pack of 2550 mAh = **4.25h - 9.18km**

Manduca G., Padovani L., Santaera G., Graziani G., Dario P., Romano D., Stefanini C. (2025, May). A Novel Underwater Robot with Carangiform Locomotion Achieved via Single Degree of Actuation and Magnetically Transmitted Traveling Wave. In 2025 IEEE International Conference on Robotics and Automation (ICRA). IEEE.





Vision-based system integration

On-board microcomputer (Raspberry Pi 4 Model B) for intelligence autonomy



Integration of vision-based models for autonomous detection and recognition of marine species



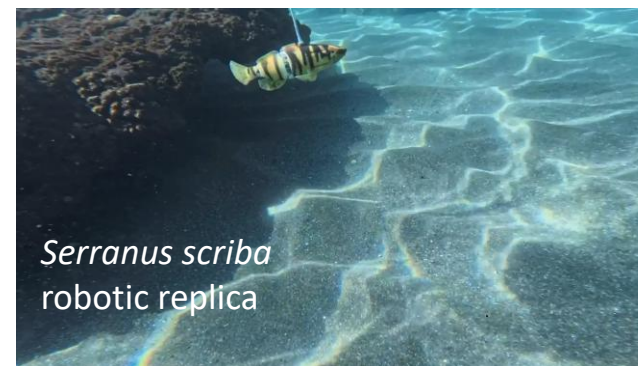


OCEAN-ROBOCTO



Prof. Donato Romano
The BioRobotics Institute
Scuola Superiore Sant'Anna

OCEAN-ROBOCTO investigates octopus-fish ethology, examining potential cognitive abilities in octopuses using fish-like robots to study interspecies communication in Australia and Italy. In this slide are presented robotic platforms designed and developed at SSSA.

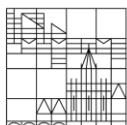


Other Partners

MAX PLANCK INSTITUTE
OF ANIMAL BEHAVIOR



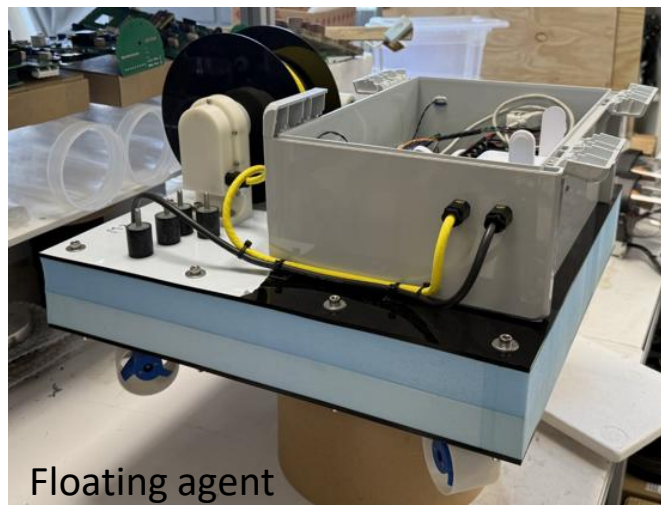
Universität
Konstanz



March 2024 – Present
Total funds awarded: 149,132 USD
National Geographic Meridian Project
OCEAN-ROBOCTO (NGS-106067R-23)



Remote control and vision



Floating agent



Plectropomus leopardus
robotic replica



November 03, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

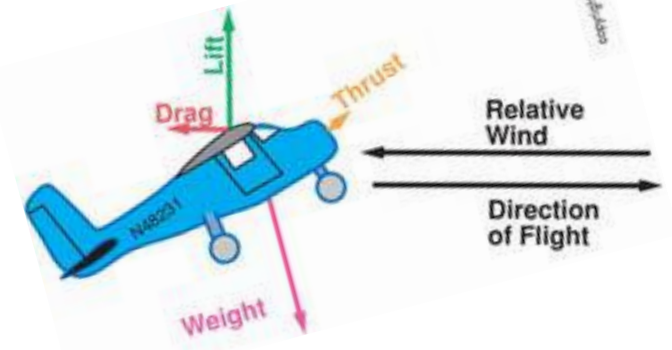
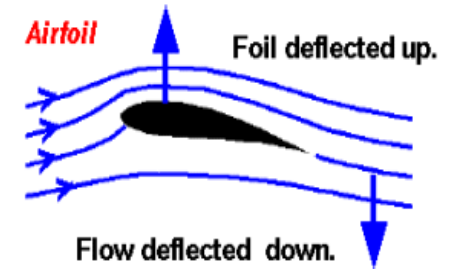
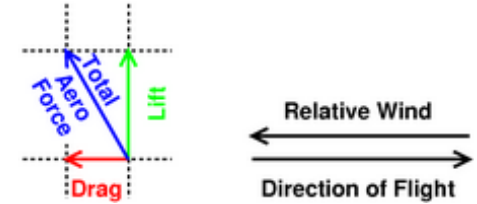
Flight

Prof. Donato Romano (donato.romano@santannapisa.it)



Aerodynamic forces

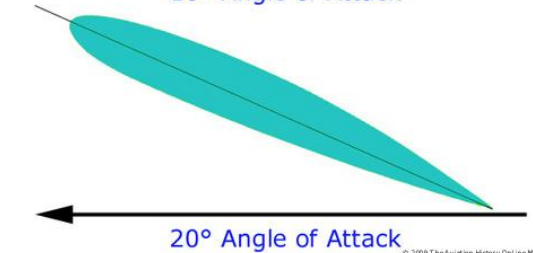
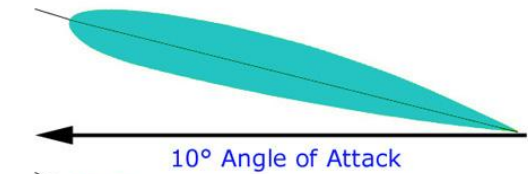
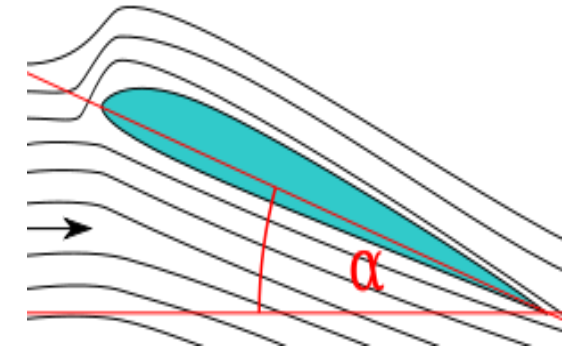
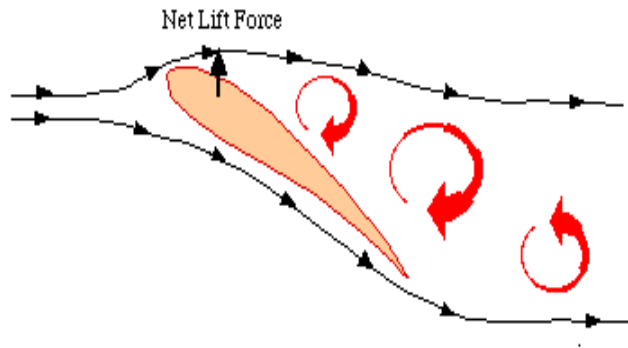
- The relative wind acting on an airfoil produces a certain amount of force which is called the *total aerodynamic force*
 - **Lift** is the component of aerodynamic force perpendicular to the relative wind.
Due to difference in pressures in upper and lower surface of wing
Upper surface of air deflects the air downwards
Lift increases with increase in angle of attack
 - **Drag** is the component of aerodynamic force parallel to the relative wind.
 - **Weight** is the force directed downward from the center of mass of the airplane towards the center of the earth. It is proportional to the mass of the airplane times the strength of the gravitational field.
 - **Thrust** is the force produced by the engine. It is directed forward along the axis of the engine.



Aerodynamic forces

Angle of attack is a term used in aerodynamics to describe the angle between the chord line of an airfoil and the vector representing the relative motion between the airfoil and the air.

- Angle between reference line of lifting body and incoming air flow
- Also called as **angle of incidence**
- **Coefficient of lift** increases with increase in angle of attack up to critical angle of attack
- After critical angle of attack, **stalling** occurs. Less lift, airflow above the wing becomes turbulent



Flight

- Gravity is a major problem for flight through the air.
 - Because it is impossible for any organism to approach the density of air, **flying animals must generate enough lift to ascend and remain airborne.**
 - Wing shape is crucial in achieving this, generating a pressure gradient that results in an upward force on the animal's body.
 - The same principle applies to airplanes, the wings of which are also airfoils.
- Other structural modifications of flying animals include
 - reduced and redistributed body weight,
 - fusiform shape, and
 - powerful flight muscles.

The AIRFOIL

An **airfoil** ([American English](#)) or **aerofoil** ([British English](#)) is the cross-sectional shape of a [wing](#), blade (of a [propeller](#), [rotor](#), or [turbine](#)), or [sail](#) (as seen in [cross-section](#)).

An airfoil-shaped body moving through a [fluid](#) produces an [aerodynamic force](#). The component of this force [perpendicular](#) to the direction of motion is called [lift](#). The component parallel to the direction of motion is called [drag](#). [Subsonic flight](#) airfoils have a characteristic shape with a rounded [leading edge](#), followed by a sharp [trailing edge](#), often with a [symmetric](#) curvature of upper and lower surfaces. Foils of similar function designed with water as the working fluid are called [hydrofoils](#).



Low-speed ULM (1 m)



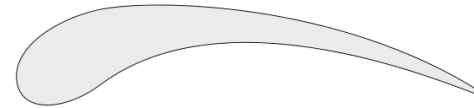
Airliner (8 m)



Propeller blade (15 cm)



Supersonic interceptor (2 m)



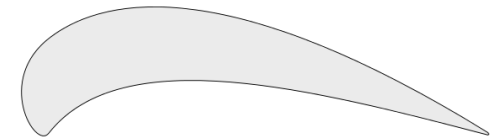
Blackbird (6 cm)



Turbofan fan blade (80 cm)



Dragonfly wing (12 mm)



Turbine blade (8 cm)



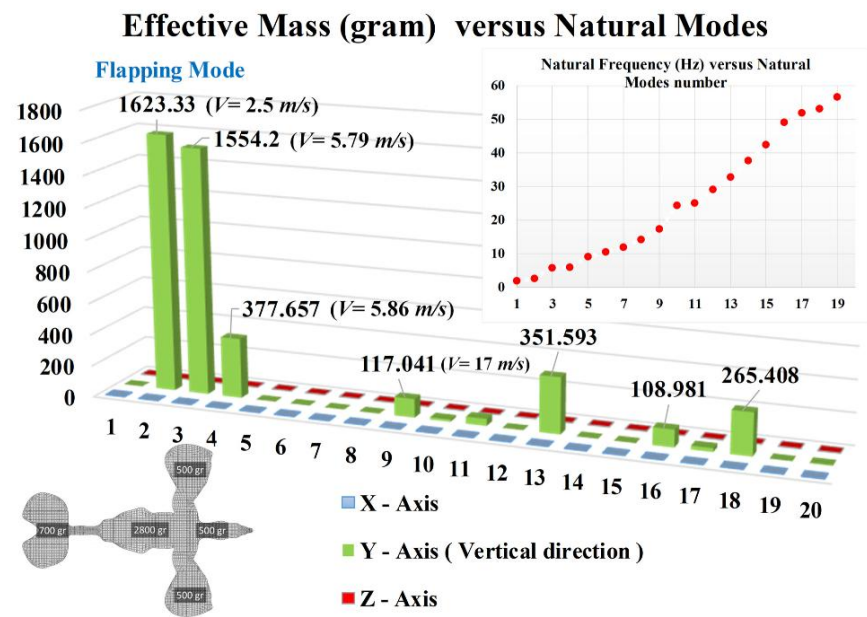
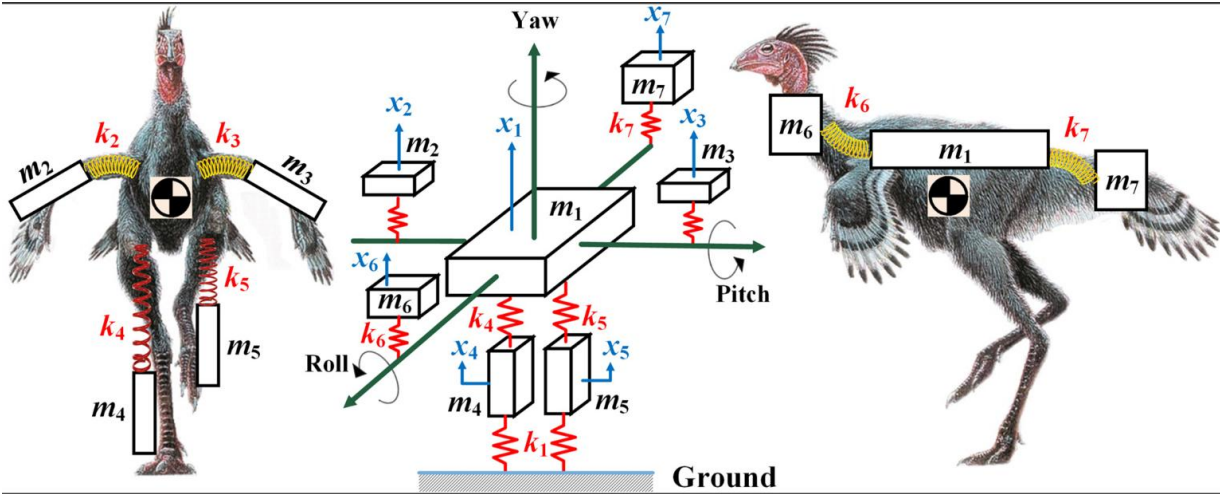
Dolphin flipper fin (10 cm)



Sailboat (3 m)

Identification of avian flapping motion from non-volant winged dinosaurs based on modal effective mass analysis

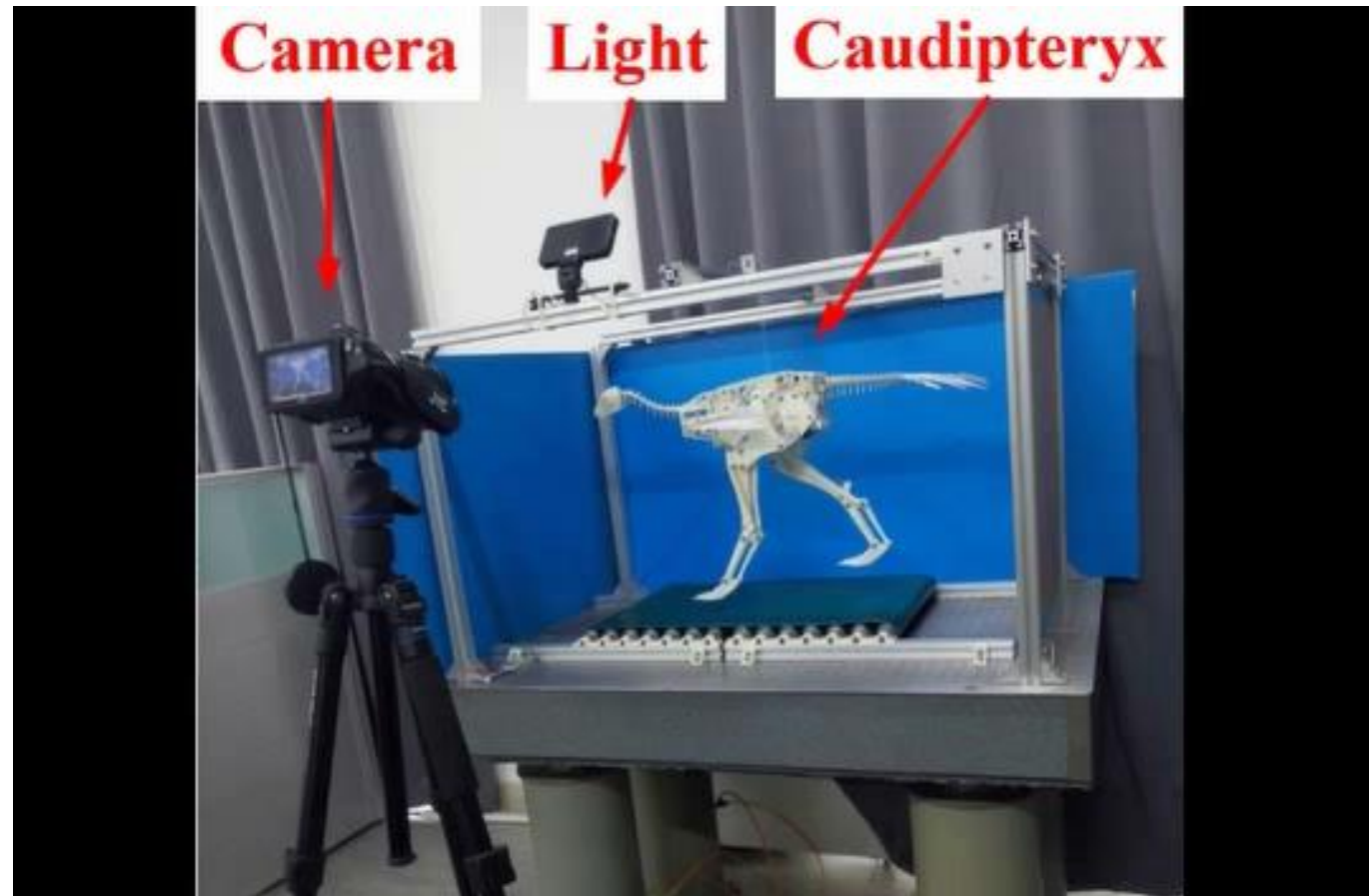
The origin of avian flight is one of the most controversial debates in Paleontology. This paper investigates the wing performance of *Caudipteryx*, the most basal non-volant dinosaur with pennaceous feathered forelimbs by using modal effective mass theory. From a mechanical standpoint, the forced vibrations excited by hindlimb locomotion stimulate the movement of wings, creating a flapping-like motion in response. **This shows that the origin of the avian flight stroke should lie in a completely natural process of active locomotion on the ground.** In this regard, flapping in the history of evolution of avian flight should have already occurred when the dinosaurs were equipped with pennaceous remiges and rectrices. The forced vibrations provided the initial training for flapping the feathered wings of theropods similar to *Caudipteryx*.



Identification of avian flapping motion from non-volant winged dinosaurs based on modal effective mass analysis

Before they evolved the ability to fly, two-legged dinosaurs may have begun to flap their wings as a passive effect of running along the ground

<https://phys.org/news/2019-05-dinosaurs-wings-evolved.html?fbclid=IwAR0O7AtGkKn6YPwxq-Tjq7sp89HgxE48rdALw7GxnAw66MWJYx2q0bXjVvM>



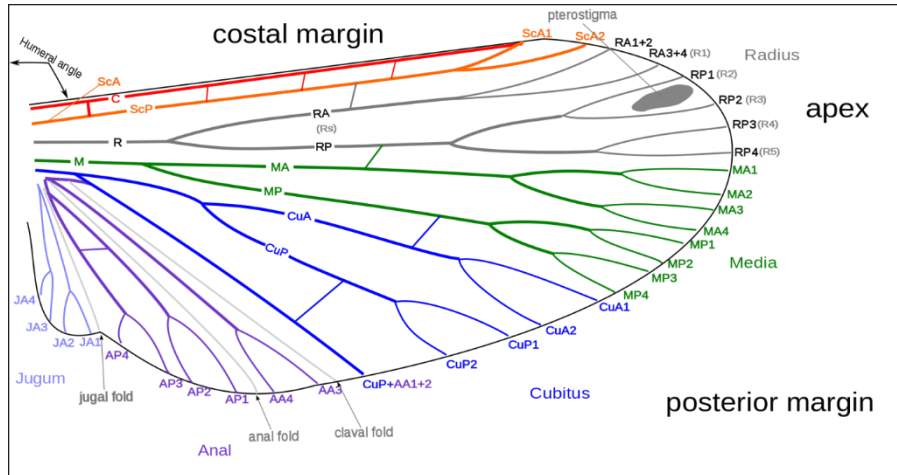
Talori, Y. S., Zhao, J. S., Liu, Y. F., Lu, W. X., Li, Z. H., & O'Connor, J. K. (2019). Identification of avian flapping motion from non-volant winged dinosaurs based on modal effective mass analysis. *PLoS computational biology*, 15(5), e1006846.

Insect wings

- **Insect wings** come in a huge variety of shapes, sizes, and appearance. However, they resemble each other in that they are passive membranous foils that depend largely on the arrangement of their supporting framework for many of their aerodynamic properties. So, during flapping flight, the wings become **distorted** by the changing forces acting on them, but in an effective and efficient way
- The **leading edge** of the wing has a much stronger and stiffer structure than the other regions of the wing which are more flexible and capable of twisting
- Most insect wings are **twisted**, like helicopter blades, with a higher angle of attack at the base. The twist is generally between 10 and 20 degrees
- Most insects control their wings by adjusting **tilt**, **stiffness**, and **flapping frequency** of the wings
- Energy is put into the wings from the root by the **powerful muscles** within the insect's thorax. The joints between wing and body are remarkable for the complex movements they allow and their ability to store and release energy appropriately at different stages in the flapping cycle
- **Typical insect flyers can attain lift forces up to** three times their weight and horizontal thrust forces up to **five times** their weight.

Insect venation

The arrangement of veins on the wings is called venation and it is extensively used in insect classification



Example of wings- small insects

Costa (C) – the leading edge of the wing

Subcosta (Sc) – second longitudinal vein (behind the costa), typically unbranched

Radius (R) – third longitudinal vein, one to five branches reach the wing margin

Media (M) – fourth longitudinal vein, one to four branches reach the wing margin

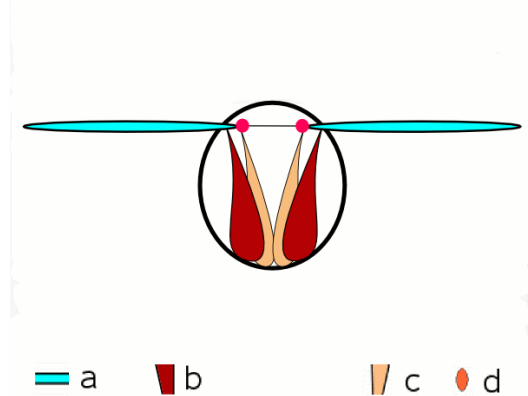
Cubitus (Cu) – fifth longitudinal vein, one to three branches reach the wing margin

Anal veins (A1, A2, A3) – unbranched veins behind the cubitus

- Small veins often found inter connecting the longitudinal veins are called cross veins
- Due to the presence of longitudinal veins and cross veins, the wing surface gets divided into a number of enclosed spaces termed cells

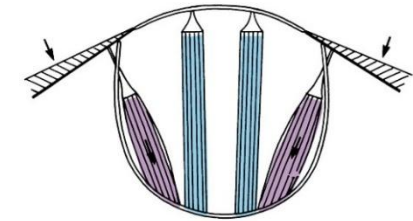
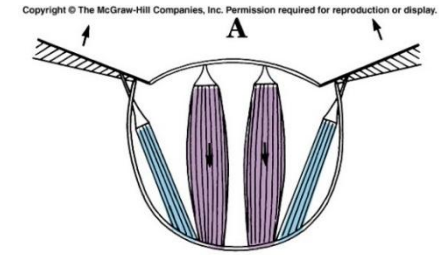
Insect Flight - Direct

Direct flight: muscles attached to wings. Large insect



Mechanism of the wing beat in insects. to: a: wings; b: axillary sclerites; c: indirect sternal the back muscles; d: indirect dorsal longitudinal muscles.

- Two living orders with direct flight muscles (dragonflies and damselflies) The wing muscles insert directly at the wing bases, which are hinged so that a small movement of the wing base downward, lifts the wing itself upward like rowing through the air
- It requires **two nerve impulses** for upward and downward stroke
- In mayflies, the hind wings are reduced, sometimes absent, and play little role in their flight, which is not particularly agile
- In odonates (dragonflies), the fore and hind wings are similar in shape and size, and operated independently, and as aerial predators evolved more advanced flight ability

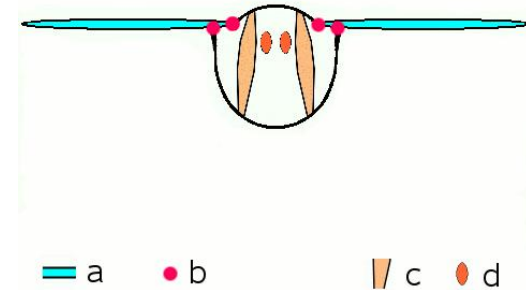


Direct and indirect flight muscles of locusts and dragonflies

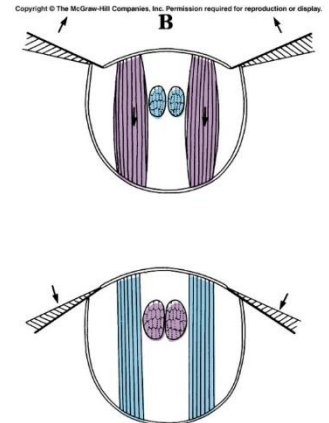
Insect Flight - Indirect

- All other living winged insects fly using a different mechanism, involving indirect flight muscles
- Indirect flight muscles allow wings to **beat faster** than neural transmission
- It corresponds with the appearance of a wing-folding mechanism, which allows Neopteran insects to fold the wings back over the abdomen when at rest
- Muscles act **to change the shape** of the exoskeleton
- The wing muscles of Paleopterans insert directly at the wing bases, which are hinged so that a small movement of the wing base downward, lifts the wing itself upward

Indirect flight: muscles make thorax oscillate in most insects: butterflies



Mechanism of the wing beat in insects. to: a: wings; b: axillary sclerites; c: indirect sternal the back muscles; d: indirect dorsal longitudinal muscles.



Indirect flight muscles of flies and midges

November 06, 2025

Master Degree in Bionics Engineering

Course of Principles of Bionics and Biorobotics Engineering

Lesson title:

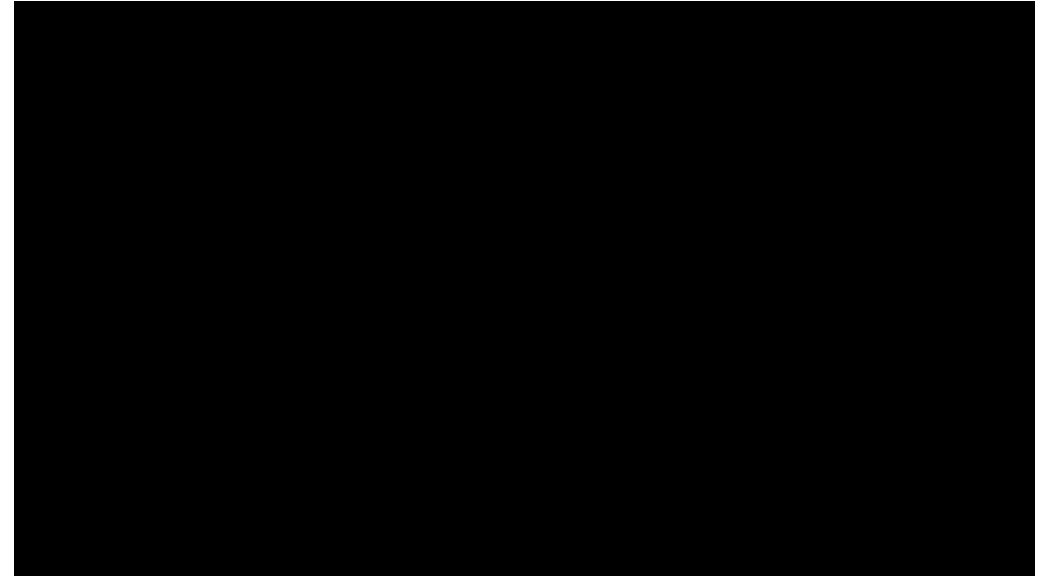
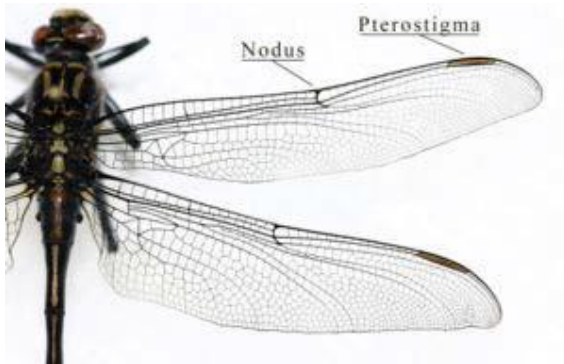
Flight

Prof. Donato Romano (donato.romano@santannapisa.it)



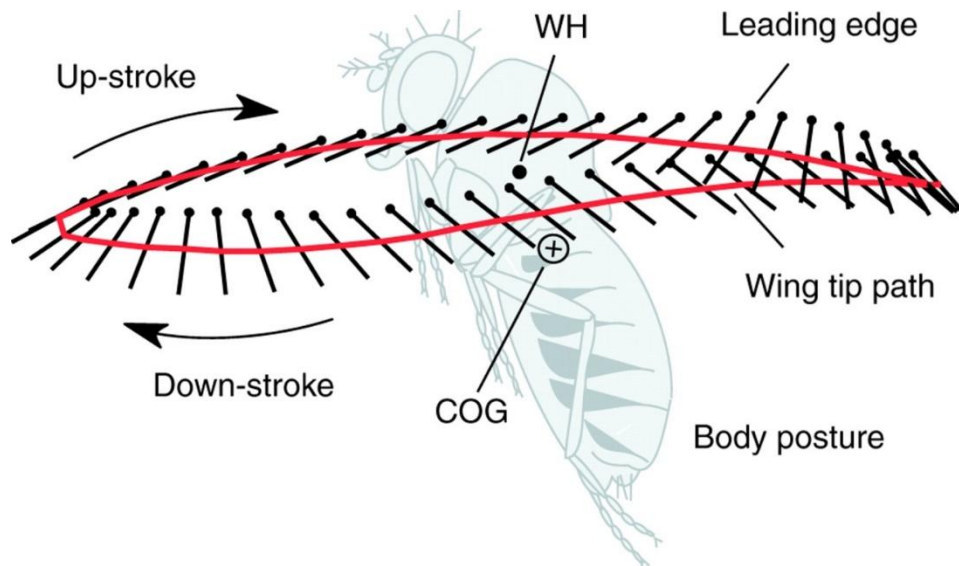
Dragonfly

- Dragonflies are very maneuverable in flight. The flight muscles can adjust **stroke frequency, amplitude, phase** between forewings and hindwings, and angle of attack independently for each of the four wings.
- Four main flying modes are recognized:
 - **counter-stroking**, in which forewings and hindwings move up and down about 180° out of phase - this pattern is generally used in cruising flight
 - **phased-stroking**, where the hindwings cycle about 90° before the forewings - this provides more acceleration for manoeuvres
 - **synchronized-stroking**, where the forewings and hindwings move in unison
 - **gliding**, where the wings are held without beating for free gliding, gaining lift in updrafts, or during mating

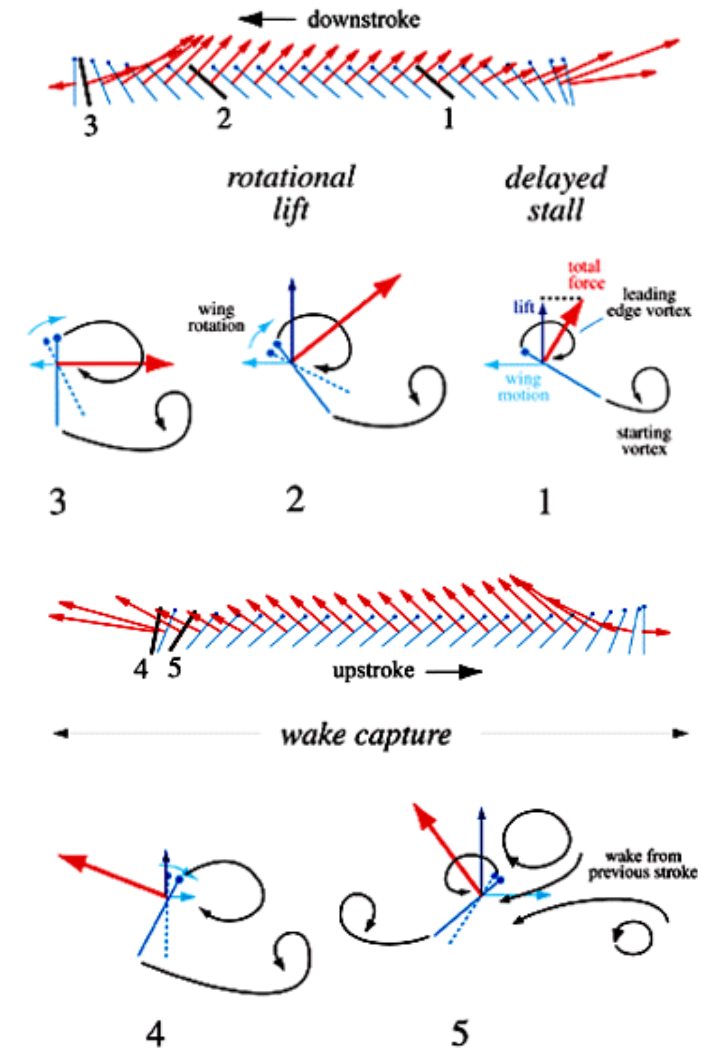


Wing Thrust and twist

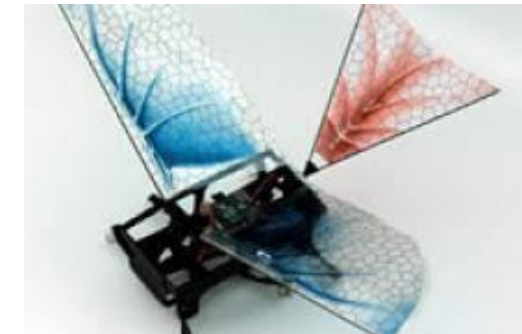
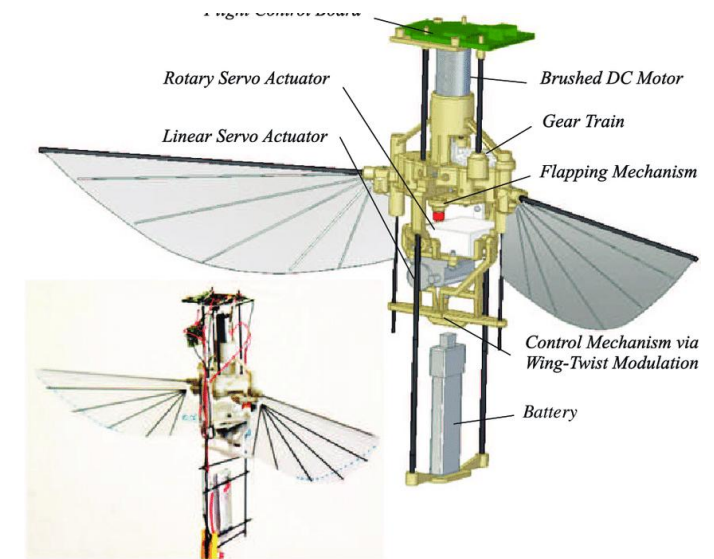
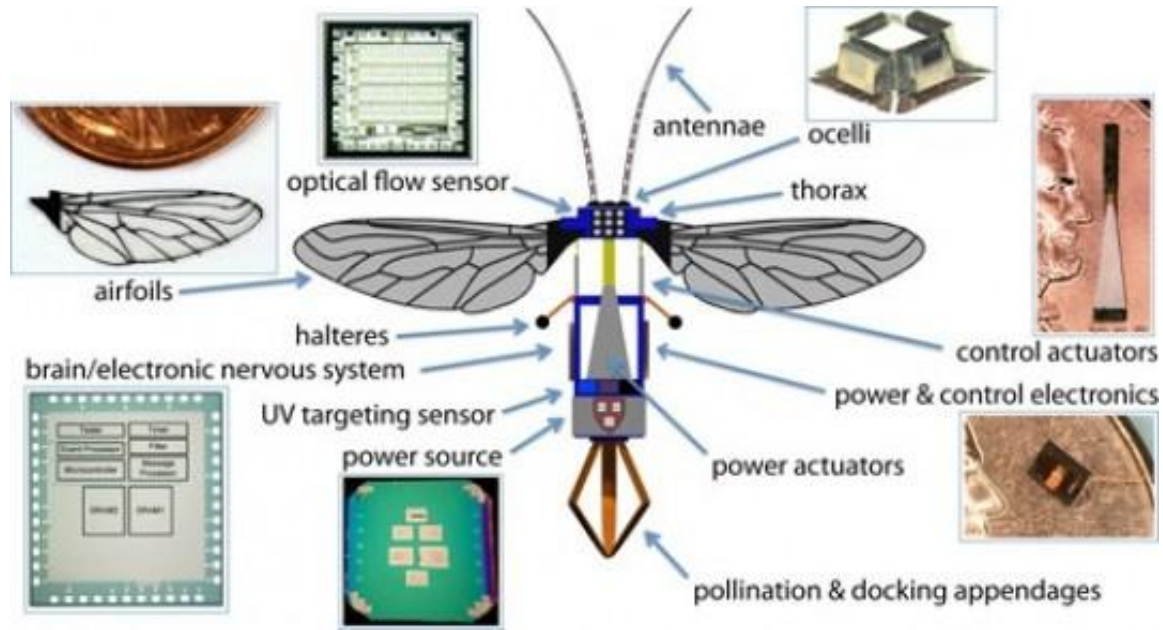
- Direct flight muscles also alter the angle of wings to **twist** the leading edge to provide **thrust**.
- Fast flight requires long, narrow wings and a strong tilt, as in dragonflies and horse flies.



Wingstroke is divided into four stages: two translational (**upstroke** and **downstroke**) and two rotational phases (**pronation** and **supination**)



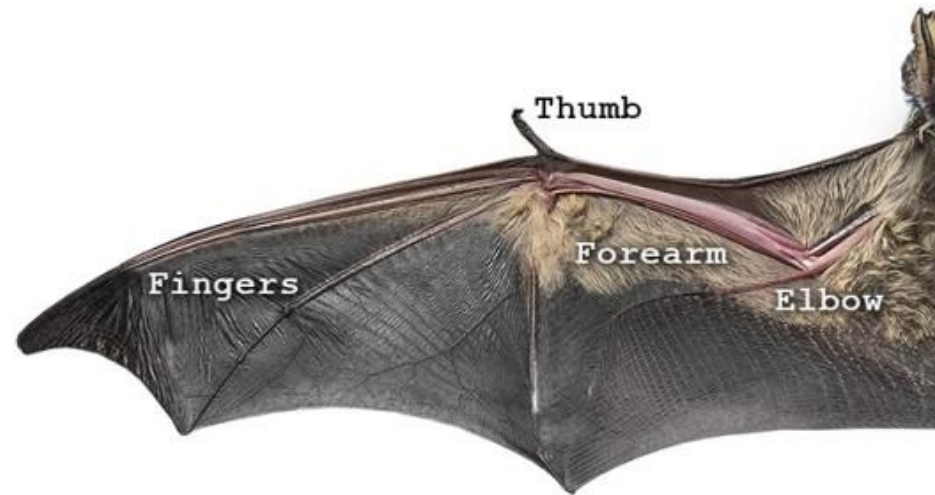
COLIBRI: A hovering flapping twin-wing robot



Bob Wood Harvard (Microrobotics Laboratory) and Ron Fearing (Biomimetic Millisystems Lab, UC Berkeley)

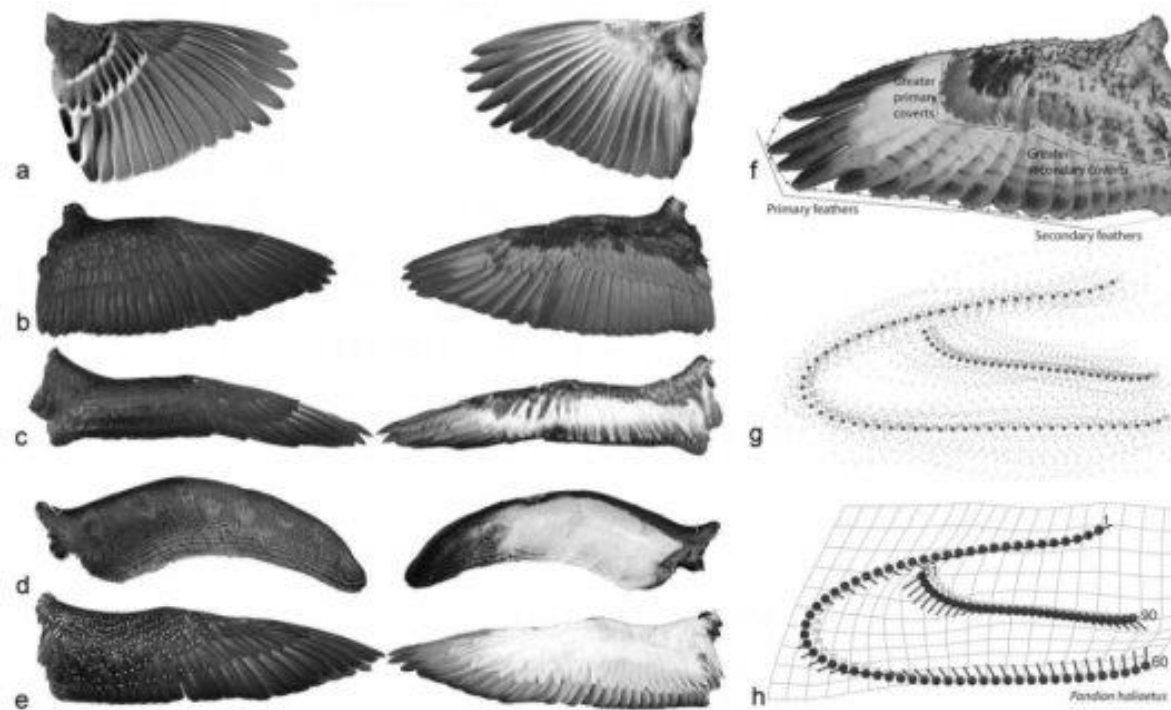
Bat's wing

- Bats' wings have a much more **dynamic geometry** than those of insects.
- The slender bony framework is jointed, and the relative positions of bones are controlled by numerous muscles **within the wing** and **attaching to it** from the body.
- Thus the repertoire of movements is much greater than for insects, both during flapping flight and when the wings are folded at rest.
- However, like insects the main area of the foil is composed of membrane, although in this case it is an **elastic and adaptable membrane** rather than relatively non-stretchable as in the insect's wing.



Bird's wing

- **Birds** go one step further than bats - instead of a membrane stretched elastically between the skeletal elements, they have numerous **feathers**, each shaped and equipped for a number of functions.
- This greatly increases the range of geometries over which an aerodynamically effective foil can be maintained.
- Although birds' wing share similar structures and composition, different species have emerged with sophisticated adaptations for different modes of flight:
 - long range
 - high speed
 - hovering
 - short take-off,



Flight Forms

Unpowered aerial locomotion

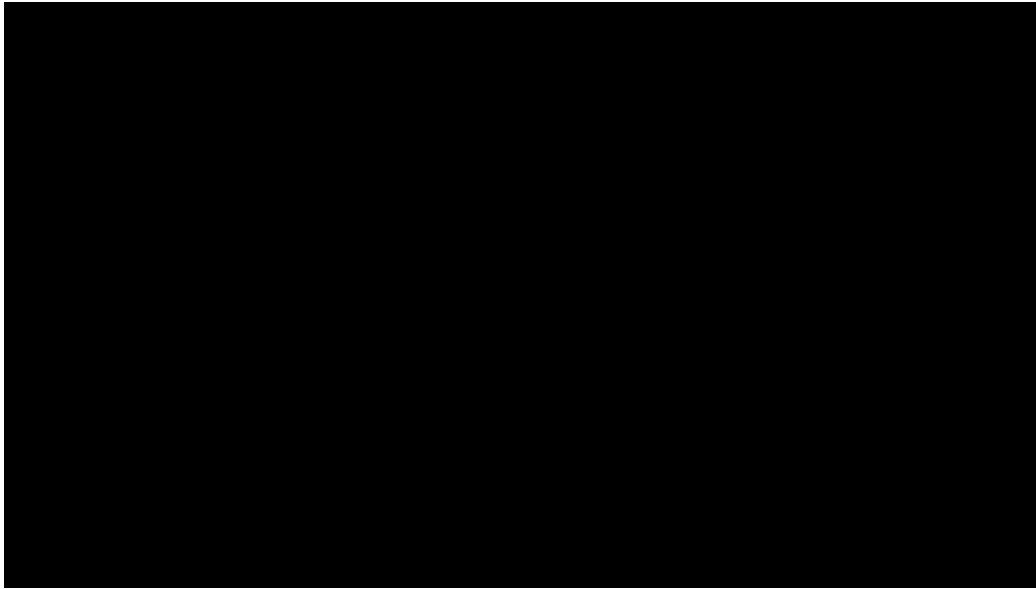
- **Falling / Diving**: Decreasing altitude under the force of gravity, using no adaptations to increase drag or provide lift
- **Parachuting**: Defined as falling at greater than 45 degrees from the horizontal with adaptations to increase drag forces. Very small animals may be carried up by the wind.
- **Gliding**: Defined as falling at less than 45 degrees from the horizontal. Lift caused by some kind of aerofoil mechanism, allowing slowly falling directed horizontal movement. Streamlined to decrease drag forces to aid aerofoil. Often some manoeuvrability in air

Flight Forms

Externally powered aerial locomotion

Powered by external aerodynamic sources of energy (wind and rising thermals, respectively)

- **Ballooning**: being carried up into the air from the aerodynamic effect on long strands of silk in the wind. Certain silk-producing arthropods, mostly small or young spiders, secrete a special light-weight gossamer silk for ballooning, sometimes traveling great distances at high altitude.
- **Soaring**: Appears similar to gliding but is actually very different, requiring specific physiological and morphological adaptations. The animal keeps aloft on rising warm air (thermals) without flapping its wings. Only large animals can be efficient soarers.



Flight Forms

Powered aerial locomotion

- **Flapping:** Flapping of wings to produce thrust. May ascend without the aid of the wind, as opposed to gliders and parachutes. Evolved four times, with different wings parameters:

Evolution
↓

- Insects
- Pterosaurs: flying dinosaur, now extinct. The wing was a membrane supported by the bony skeleton of the arm, forearm and hand
- Birds: the bones of the hand support the primary feathers and the bones of the forearm support the secondaries. A small alula is supported in the 'thumb' position
- Bats: an elastic membrane is supported by the skeletal elements. The long slender digits are distinctive

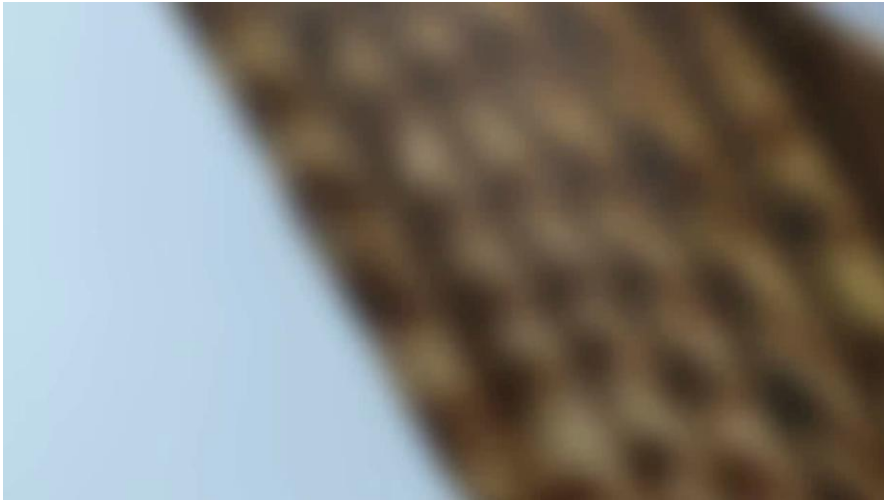
1. pterosaur (Pterosauria)
2. bat (Chiroptera)
3. bird (Aves)



Chasing away nuisance birds

The Robird® version of the peregrine is just as intimidating to birds as the real deal. After a couple of flights, the bird population understands that they are living in a dangerous hunting territory, and will take their business elsewhere. By triggering the instinct of birds, through the combination of silhouette and wing movement, chasing off birds becomes fully controllable

The logo for CLEAR FLIGHT SOLUTIONS features a stylized graphic of two birds in flight above the company name. The text "CLEAR FLIGHT" is in a large, bold, sans-serif font, and "SOLUTIONS" is in a smaller, all-caps, sans-serif font below it.





Robird

By Gerrit Adriaan Folkertsma, Wessel Straatman, Nico Nijenhuis, Cornelis Henricus Venner, and Stefano Stramigioli

The danger that aviation poses to birds, and, in turn, to itself, necessitates the relocation of birds around airports... **Robirds** are robotic birds that closely resemble natural birds, both in appearance and in flight behavior... Its only downtime is the changing of batteries.... It is a perfect **nature-inspired and nature-friendly** means of bird control for **aviation safety**

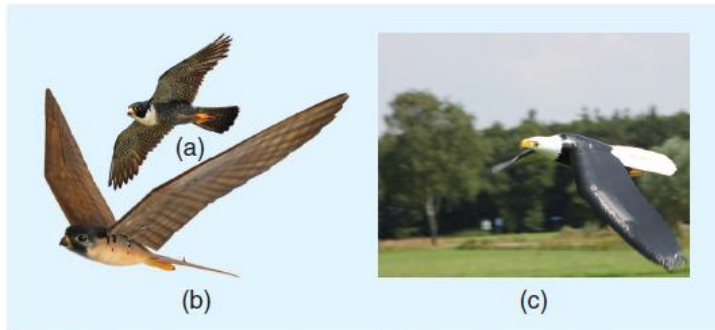


Figure 1. (a) A peregrine falcon. (Photo courtesy of Julien Baudat-Franceschi.) (b) The Robird peregrine falcon model and (c) the bald eagle model.

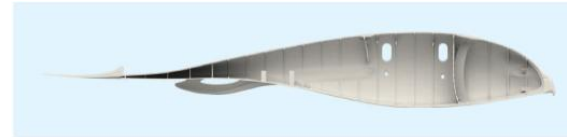


Figure 3. The hull of the Robird.

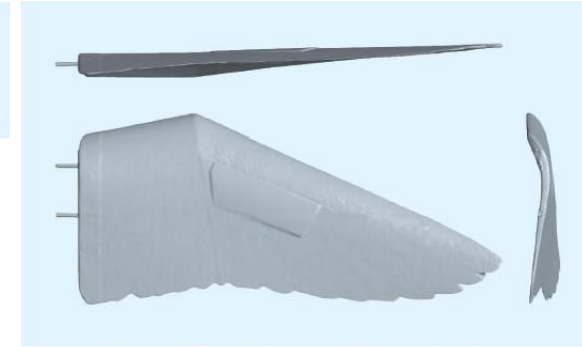


Figure 4. The wing of the Robird. Note the spoiler—the trapezoidal patch in the center—used for rolling maneuvers and the two spars used for driving the wing.

	Bald Eagle		Peregrine Falcon	
	Bird	Robird	Bird	Robird
Mass	3.0–6.3 kg	2.1 kg	0.6–1.3 kg	0.73 kg
Body length	70–102 cm	88 cm	39–50 cm	56 cm
Wingspan	1.8–2.3 m	1.76 m	0.95–1.15 m	1.12 m
Flight speed	15–20 m/s	18 m/s	18–25 m/s	16 m/s



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Fixed-Wing Aerodynamics

In fixed-wing flight mode, the aerodynamics of the Robird are essentially equal to those of a glider. A glider's performance can be expressed by its glide number, the maximum lift/drag ratio [11]

$$(L/D)_{\max} = \frac{1}{2} \sqrt{\frac{\pi e A R}{c_{D,0}}}, \quad (1)$$

where AR is the glider's aspect ratio, b^2/S , where b is the total wingspan and S the wing area plus the area of the body directly between the wings; $c_{D,0}$ is the drag coefficient at zero lift ($c_L = 0$); and e is the Oswald span efficiency factor [12], [13].

The values for the Robird model (see Table 2) give a glide ratio of 8.3. This implies that it can cover a forward distance of 8.3 m for every 1-m loss in altitude. For comparison, the glide ratio for a glider plane is over 40, and that of a Boeing 747 is approximately 14.

the real falcon has a similar low-glide ratio of 10 [14]. The Robird's poor performance as a glider is partly due to the locking mechanism, which locks the wings when moving downward with a negative angle of attack. Also, the drag coefficient $c_{D,0}$ is very high—the wing shape of both a falcon and Robird is clearly optimized for fast flapping-wing flight.

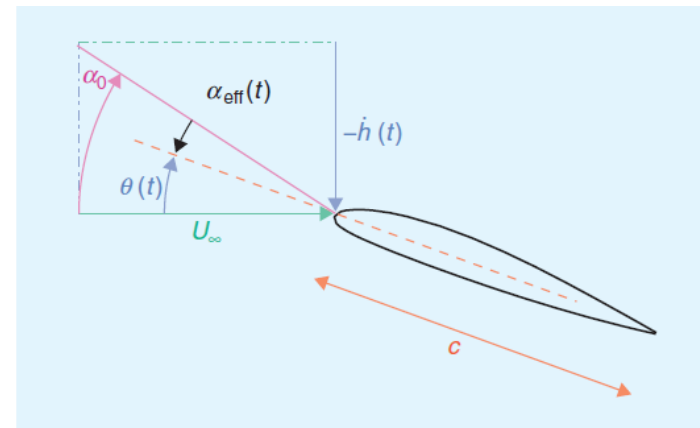


Figure 5. A 2-D cross-section of the wing: The plunging velocity ($\dot{h}(t)$) and flight velocity U_∞ result in an angle-of-attack α_0 , which is reduced due to pitching ($\theta(t)$) to the effective angle-of-attack $\alpha_{\text{eff}}(t)$. (The chord length of the wing is c .)

Take-off

- **Take-off** can be one of the most energetically demanding aspects of flight, as the bird needs to generate enough airflow under the wing to create lift.
 - In small birds a **jump up** will suffice, while for larger birds this is simply not possible.
 - In this situation, birds need to take a **run up** in order to generate the airflow to take off.
 - Large birds often simplify take off by **facing into the wind**, and, if they can, **perching on a branch** or cliff so that all they need to do is drop off into the air.



Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”

ROBOTICS

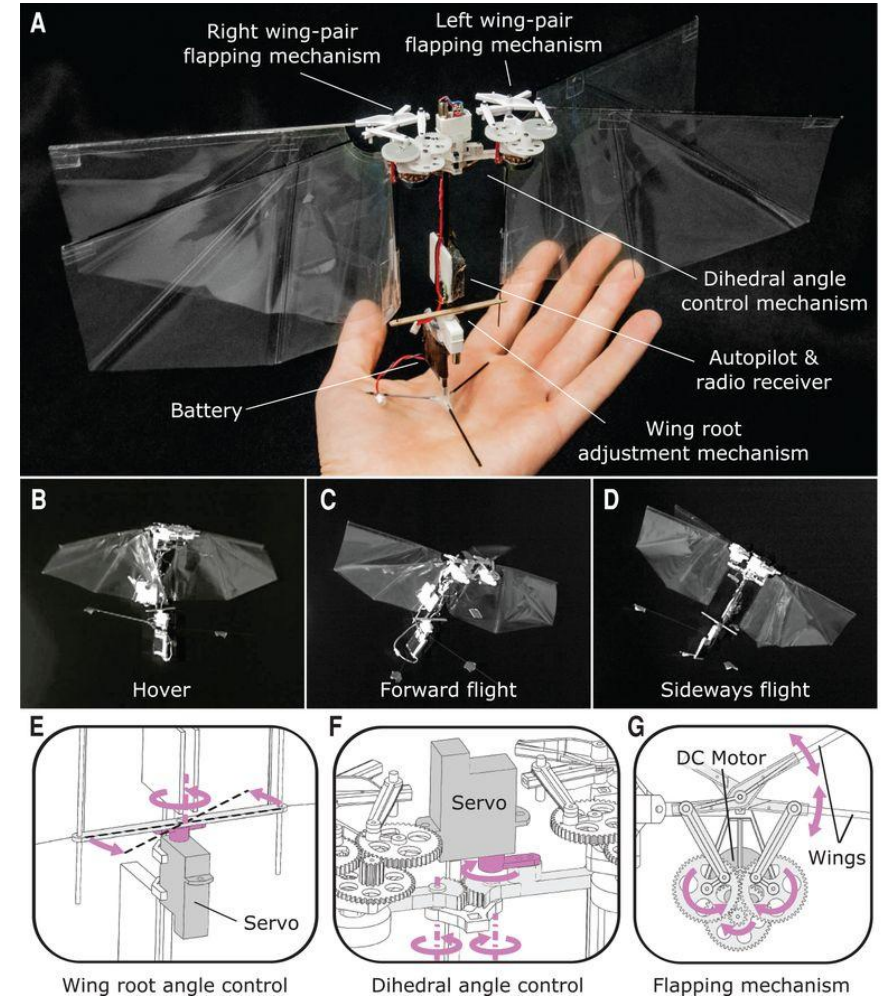
A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns

Matěj Karásek^{1*}, Florian T. Muijres², Christophe De Wagter¹,
Bart D. W. Remes¹, Guido C. H. E. de Croon¹

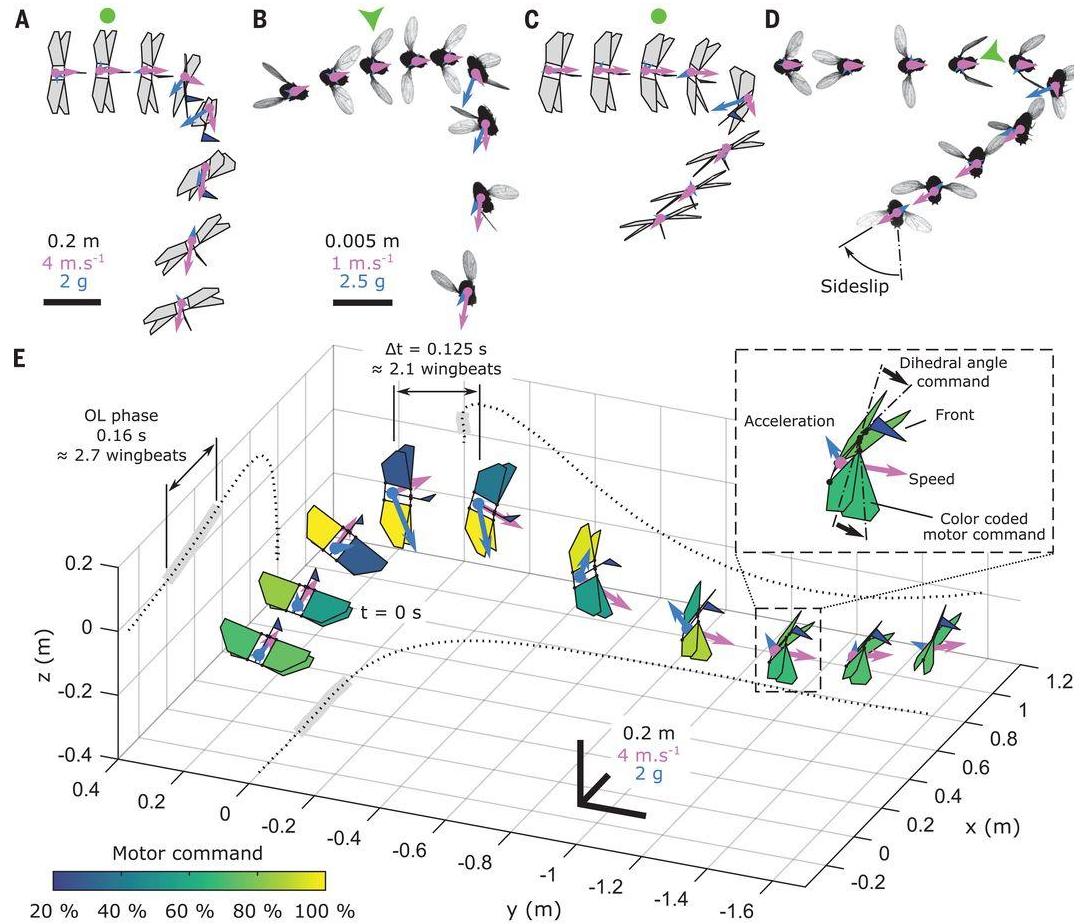
¹Micro Air Vehicle Laboratory, Control and Simulation, Delft University of Technology, Delft, Netherlands. ²Experimental Zoology Group, Wageningen University and Research, Wageningen, Netherlands.

*Corresponding author. Email: m.karasek@tudelft.nl

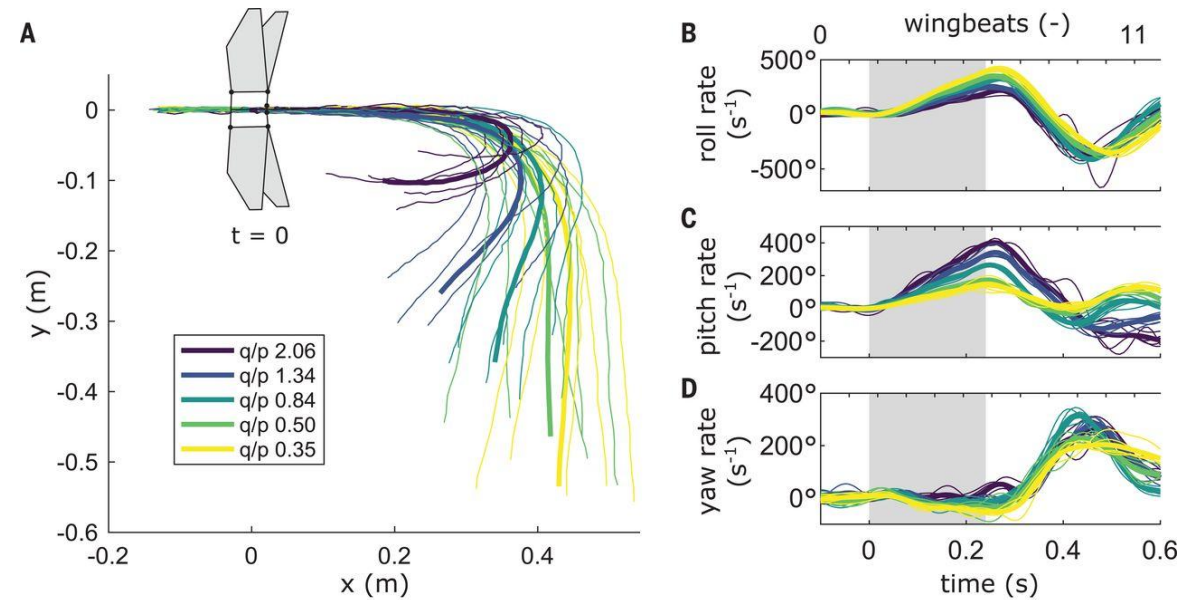
Karásek *et al.*, *Science* **361**, 1089–1094 (2018) 14 September 2018



Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”



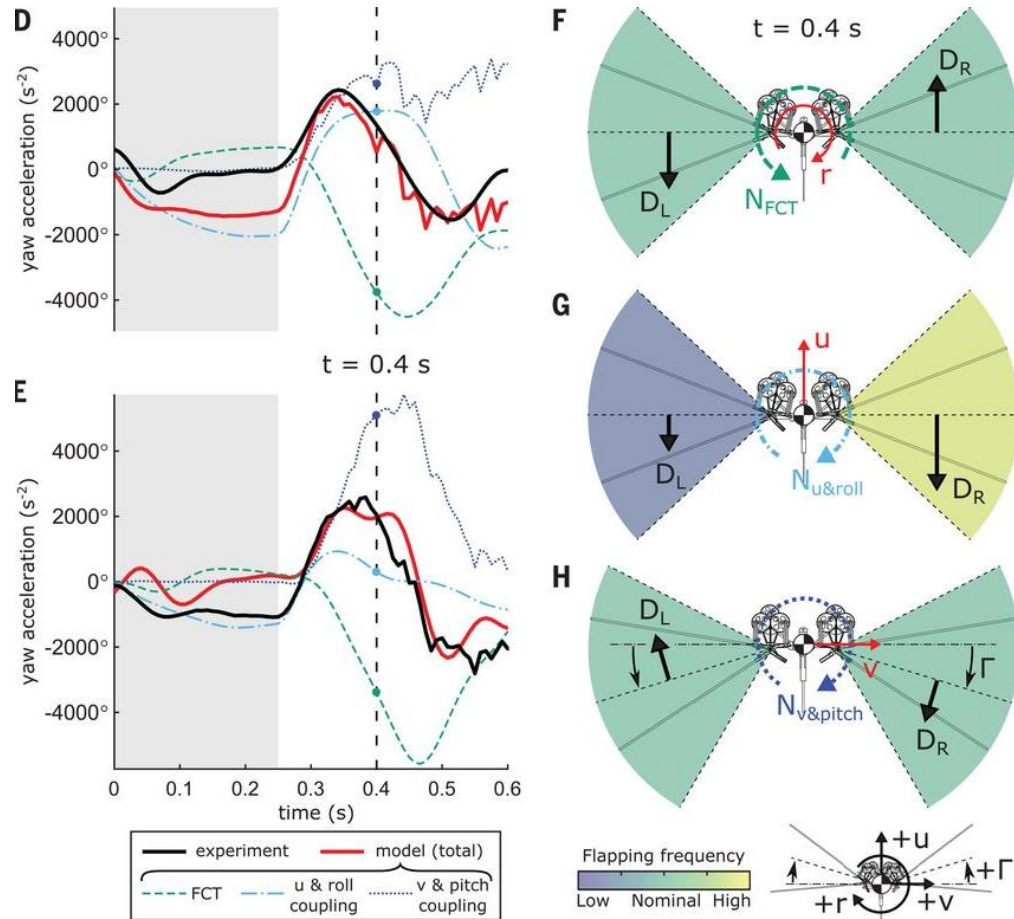
The robot mimics rapid banked turns observed in escaping fruit flies.



The turn angle during the banked turn can be controlled by varying the ratio of pitch-to-roll torque commands

M. Karásek et al. A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns. *Science*. 361: 1089-1094 (2018)

Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”



The experimental results obtained with the robot supported the **hypothesis** that fruit flies and other insects do not actively control yaw throughout evasive maneuvers, but instead use the translation-induced coupled yaw torque (equation below) to rotate their body in the direction of the banked turn.

$$N = \frac{\dot{r}}{I_{zz}} = -b\Phi R[(f_L + f_R)Rr + (f_L - f_R)u + (f_L + f_R)\Gamma v]$$

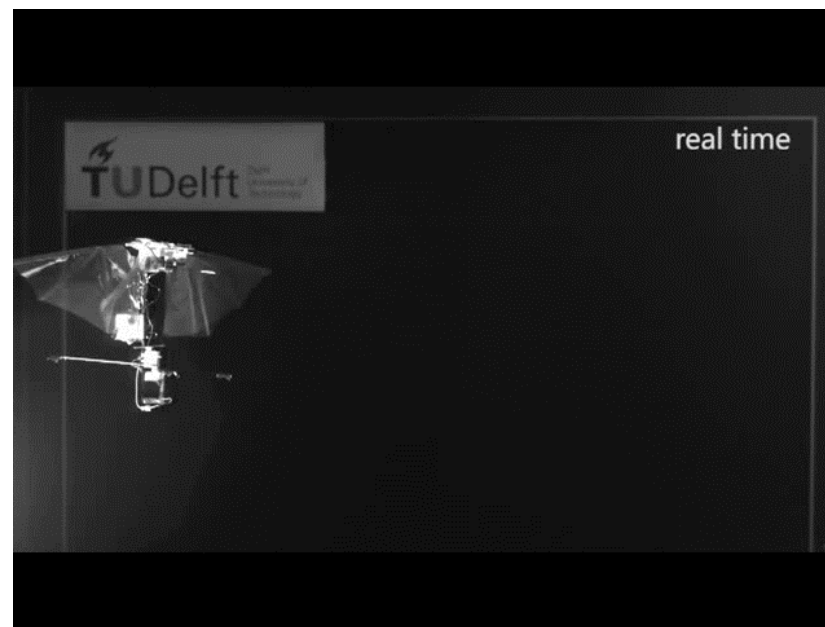
where N is yaw torque; $\dot{r}$ is yaw acceleration; I_{zz} is the moment of inertia around the vertical body axis; b is the linear damping coefficient due to flapping counterforce (22); Φ is flapping amplitude; R is wing length; f_L and f_R are flapping frequency of the left and the right wing, respectively; r is yaw rate; u and v are the forward and sideways velocity of the robot, respectively; and Γ is the wing dihedral angle.

M. Karásek et al. A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns. *Science*. 361: 1089-1094 (2018)

Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”



Robot prototype in hovering flight, whereby the autopilot maintains stable flight of the inherently unstable flight platform.



Robot prototype during the pitch maneuver. During the maneuver, the robot transitioned from hover to forward flight by rapidly pitching forward, and then pitched back again to return to a hover condition.



Robot prototype during the roll maneuver. During the maneuver, the robot transitioned from hover to sideways flight by rapidly rolling sideways, and then roll back again to return to a hover condition.

M. Karásek et al. A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns. *Science*. 361: 1089-1094 (2018)

Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”



Robot prototype rapidly accelerating from hover to fast forward flight.



Robot prototype making a 360° roll flip maneuver.

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Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”



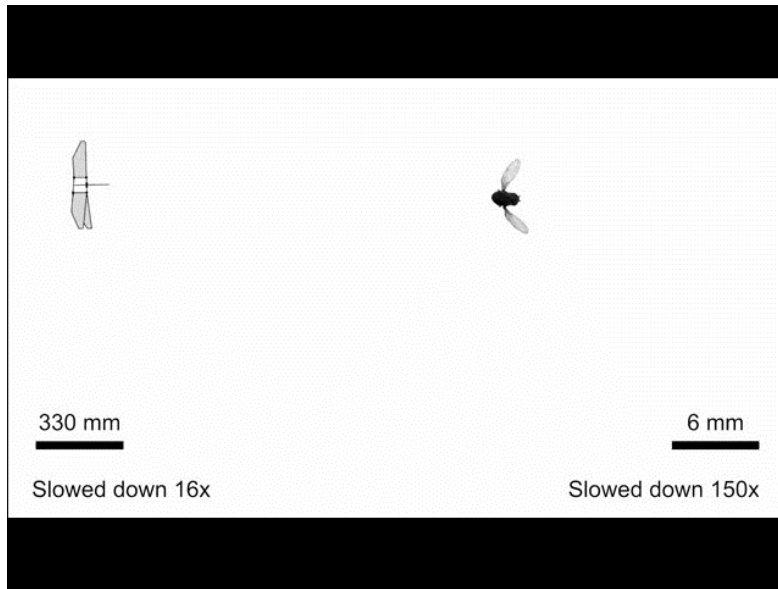
Robot prototype making 360° pitch flip maneuver with barrel rolls and loopings.



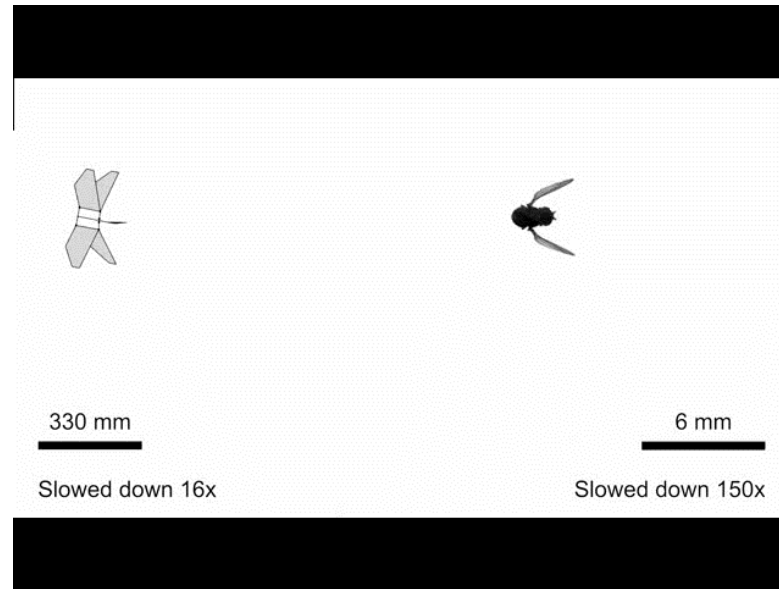
Robot prototype making a fly inspired rapid banked turn.

M. Karásek et al. A tailless aerial robotic flapper reveals that flies use torque coupling in rapid banked turns. *Science*. 361: 1089-1094 (2018)

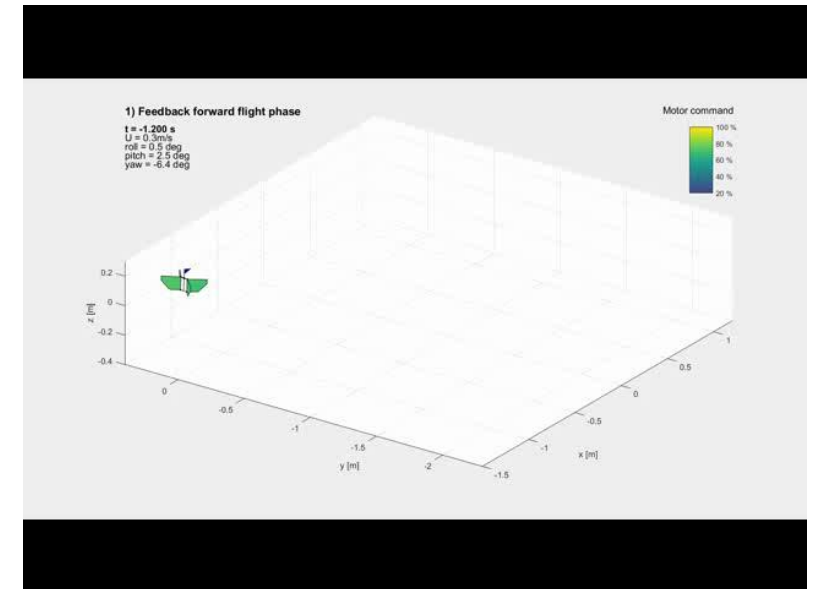
Aerial robotic flapper: a paradigmatic example of “Science for Robotics and Robotics for Science”



Animation of the comparison between the fly-inspired rapid banked turn and the original maneuver by the fruit fly.



Animation of the comparison between the fly-inspired rapid banked turn and the original maneuver by the fruit fly (close view).



Animation of the fly-inspired rapid banked turn. The movie is slowed down 6 times. The three-dimensional trajectory is shown by the magenta dotted line, and the trajectory projections are shown as black dotted lines.